

Bapuji Educational Association (Regd)

BAPUJI INSTITUTE OF HI-TECH EDUCATION

(Affiliated to Davangere University, Approved by AICTE New Delhi & Govt. of Karnataka)

DAVANGERE – 577 004

BACHELOR OF COMPUTER APPLICATION



LAB MANUAL

ANDROID PROGRAMMING

III BCA-V SEM

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PART A

1. Create an android application to display DAVANGERE UNIVERSITY as Text Message by using Button.
2. Create an android application to call different activities by using Implicit and Explicit Intents.
3. Create an android application to select items from given list by using Auto Complete Text View (ACTV).
4. Create an android application to display dropdown menu items and pick one item by using Spinner Component.
5. Create an android application to display internal storage data using Array Adapter.
6. Create an android application to display internal storage data in vertical format by using Custom Adapter.
7. Create an android application to display WhatsApp videos in grid view by using Custom Adapter.
8. Create an android application to display webpage by using Web view Component.
9. Create an android application to display different webpages in fragments by using Fragments Component.
10. Create an android application to store the data by using Shared Preferences.
11. Create an android application to demonstrate concept of SQLite Database storage method
12. Write an android program to develop Video view application

PART B

1. Create an android application to perform different types of operations (Send SMS, Making call and sending email) by using Telephony app.
2. Write an android program to develop Media player application.
3. Create an android application to display current location on Google maps by using Google-Maps Service.
4. Write an android program to develop Audio Recording application
5. Write an android program to develop Video Recording application.
6. Write an android program to develop Camera and Gallery applications.
7. Create an android application to get latitude and longitude value by using Location Service.
8. Create an android application to get the Bluetooth devices and list of devices using Bluetooth and Vibrator Service.
9. Create an android application to get the notifications on Notification Bar by Using Notification Service.
10. Create an android application to display available Wi-Fi devices and Paired Wi-Fi devices by using Wi-Fi Service.
11. Create an android application to display X, Y Sensor values by using Sensor Service.
12. Create an android application to get the System Announcements by using Broadcast Receiver.

To create a new instant app project in Android Studio, do the following:

1. Install Android Studio

- Download and install Android Studio.
- During installation, also install the Android SDK and Emulator.

2. Create a New Project

1. Open Android Studio.
2. Click **New Project**.
3. Select **Empty Views Activity**.
4. Enter:
 - Application Name
 - Package Name
 - Language: Java or Kotlin
 - Minimum SDK
5. Click **Finish**.

3. Understand Project Structure

- **MainActivity.java / MainActivity.kt** – Application logic.
- **activity_main.xml** – User Interface design.
- **AndroidManifest.xml** – Permissions and app configuration.
- **res/** – Resources (layouts, images, strings).

4. Design the User Interface

Open `activity_main.xml` and drag UI components:

- TextView
- EditText
- Button
- ImageView

5. Write Java Code

Open `MainActivity.java`:

6. Run the Application

- Connect an Android phone with USB Debugging enabled **or**
- Create an Android Emulator.
- Click the **Run ►** button.

1. Create an android application to display DAVANGERE UNIVERSITY as Text Message by using Button.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp">

    <Button
        android:id="@+id/btnShow"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show Message" />

    <TextView
        android:id="@+id/txtMessage"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:textColor="#26FF00"
        android:textSize="22sp" />

</LinearLayout>
```

MainActivity.java

```
package com.example.BIHE;

import android.app.Activity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;

public class MainActivity extends Activity {

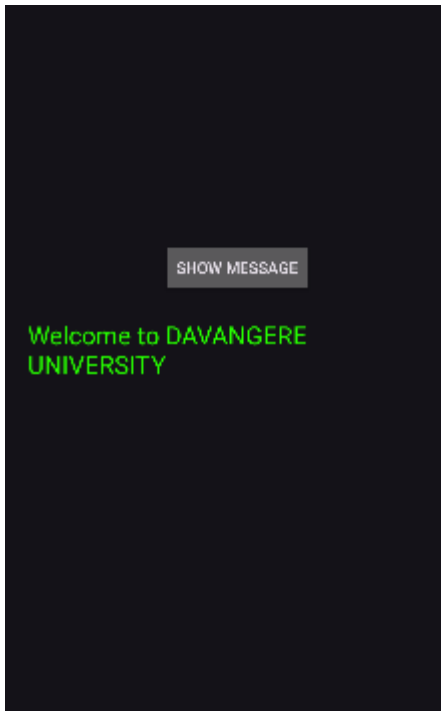
    Button btnShow;
    TextView txtMessage;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnShow = findViewById(R.id.btnShow);
        txtMessage = findViewById(R.id.txtMessage);

        btnShow.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                txtMessage.setText("Welcome to DAVANGERE UNIVERSITY");
            }
        });
    }
}
```

Output:



2. Create an android application to call different activities by using Implicit and Explicit

Intents.

1. activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <Button
        android:id="@+id/btnExplicit"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Explicit Intent"/>

    <Button
        android:id="@+id/btnImplicit"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Implicit Intent"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

2.activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Welcome to Second Activity"
        android:textSize="24sp"/>

</LinearLayout>
```

3.AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <application
        android:allowBackup="true"
        android:label="IntentDemo"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar">

        <activity android:name=".SecondActivity"/>

        <activity android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
                <category android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>

    </application>

</manifest>
```

4.MainActivity.java

```
package com.example.BIHE;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnExplicit, btnImplicit;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnExplicit = findViewById(R.id.btnExplicit);
        btnImplicit = findViewById(R.id.btnImplicit);

        // Explicit Intent
        btnExplicit.setOnClickListener(v -> {
            Intent i = new Intent(MainActivity.this,
                SecondActivity.class);
            startActivity(i);
        });

        // Implicit Intent
        btnImplicit.setOnClickListener(v -> {
            Intent i = new Intent(Intent.ACTION_VIEW,
                Uri.parse("https://www.google.com"));
            startActivity(i);
        });
    }
}
```

5.SecondActivity.java

```
package com.example.BIHE;

import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {

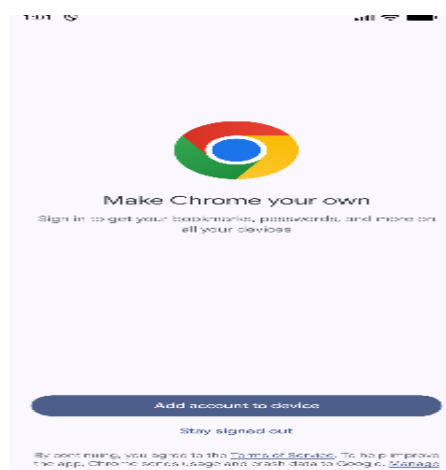
    @Override
```

```
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_second);
}
}
```

Output:



Clicking Explicit Intent → Opens SecondActivity.
 Clicking Implicit Intent → Opens Google in Browser.



3. Create an android application to select items from given list by using Auto Complete Text View (ACTV).

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="@string/enter_country"
        android:textSize="30sp"
        android:textColor="#FF0000"/>

    <AutoCompleteTextView
        android:id="@+id/actv"
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:hint="@string/type_country"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.BIHE;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.AutoCompleteTextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    AutoCompleteTextView actv;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        actv = findViewById(R.id.actv);

        String[] values = getResources().getStringArray(R.array.country);

        ArrayAdapter<String> adapter =
            new ArrayAdapter<>(
                this,
                android.R.layout.simple_dropdown_item_1line,
                values);

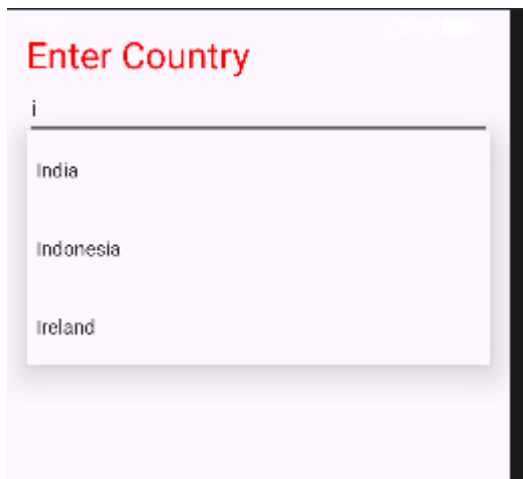
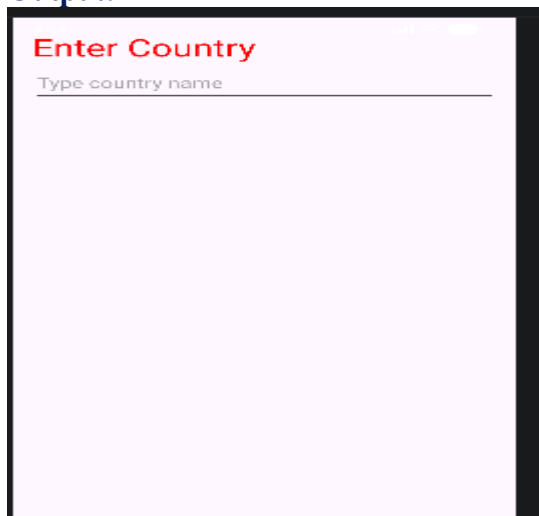
        actv.setAdapter(adapter);
        actv.setThreshold(1);
    }
}
```

strings.xml

```
<resources>
    <string name="app_name">ACTCApp</string>
    <string name="enter_country">Enter Country</string>
    <string name="type_country">Type country name</string>

    <string-array name="country">
        <item>India</item>
        <item>Indonesia</item>
        <item>Ireland</item>
        <item>Sri Lanka</item>
        <item>South Africa</item>
        <item>Pakistan</item>
        <item>Bangladesh</item>
        <item>Brazil</item>
    </string-array>
</resources>
```

Output:



4. Create an android application to display dropdown menu items and pick one item by using Spinner Component.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Select Country"
        android:textSize="30sp"/>

    <Spinner
        android:id="@+id/sp1"
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:entries="@array/countries"/>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Select Gender"
        android:textSize="30sp"
        android:layout_marginTop="20dp"/>

    <Spinner
```

```

    android:id="@+id/sp2"
    android:layout_width="match_parent"
    android:layout_height="48dp"/>

```

```
</LinearLayout>
```

MainActivity.java

```

package com.example.BIHE;

import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.AdapterView.OnItemClickListener;
import android.widget.ArrayAdapter;
import android.widget.Spinner;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;
import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Spinner sp1 = findViewById(R.id.sp1);
        sp1.setOnItemClickListener(new AdapterView.OnItemClickListener() {

            @Override
            public void onItemClick(AdapterView<?> parent,
                                    View view,
                                    int position,
                                    long id) {

                if (position > 0) {
                    Toast.makeText(
                        MainActivity.this,
                        sp1.getSelectedItem().toString(),

```

```

        Toast.LENGTH_LONG
    ).show();
    }
}

@Override
public void onNothingSelected(AdapterView<?> parent) {
    }
});

Spinner sp2 = findViewById(R.id.sp2);
ArrayList<String> list = new ArrayList<>();
list.add("Select");
list.add("Male");
list.add("Female");
list.add("TransGender");

ArrayAdapter<String> adapter =
    new ArrayAdapter<>(
        this,
        android.R.layout.simple_spinner_dropdown_item,
        list);

sp2.setAdapter(adapter);
}
}

```

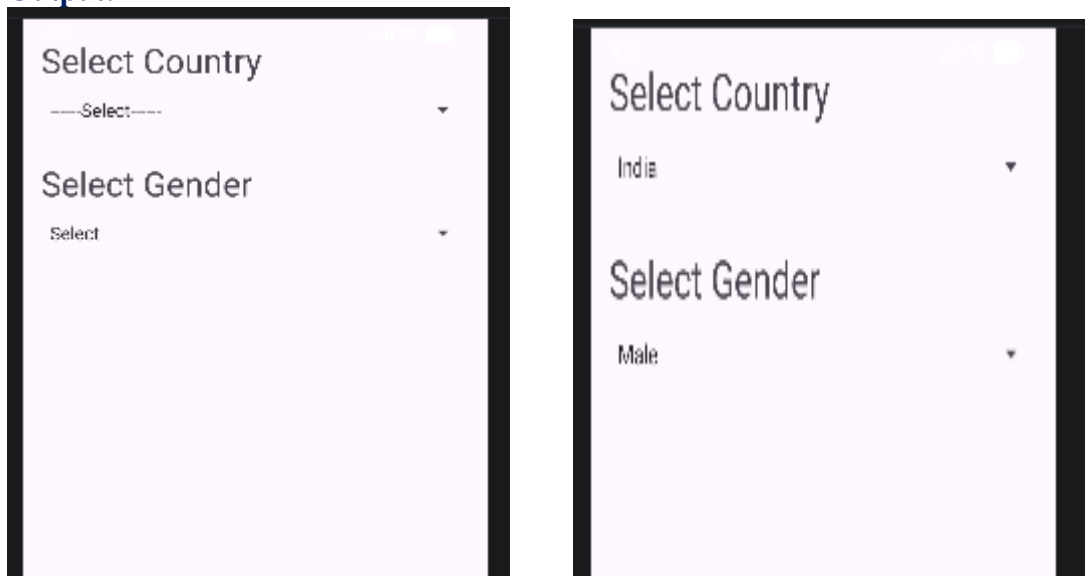
strings.xml

```
<resources>

    <string name="app_name">SPINNERApp</string>

    <string-array name="countries">
        <item>-----Select-----</item>
        <item>India</item>
        <item>Ireland</item>
        <item>Australia</item>
        <item>China</item>
        <item>Sri Lanka</item>
        <item>Bangladesh</item>
    </string-array>

</resources>
```

Output:

5. Create an android application to display internal storage data using Array Adapter.**activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <TextView

        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Internal Storage Files"
        android:textSize="24sp"
        android:gravity="center"/>

    <ListView

        android:id="@+id/lview"
        android:layout_width="match_parent"
        android:layout_height="match_parent"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.BIHE;

import android.os.Bundle;
import android.util.Log;
import android.widget.ArrayAdapter;
import android.widget.ListView;

import androidx.appcompat.app.AppCompatActivity;

import java.io.File;

public class MainActivity extends AppCompatActivity {

    @Override
```

```
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    // Display internal storage path in Logcat
    Log.d("PATH", getFilesDir().getAbsolutePath());

    ListView lview = findViewById(R.id.lview);

    try {
        openFileOutput("File1.txt", MODE_PRIVATE).close();
        openFileOutput("File2.txt", MODE_PRIVATE).close();
        openFileOutput("File3.txt", MODE_PRIVATE).close();
    } catch (Exception e) {
        e.printStackTrace();
    }

    File dir = getFilesDir();
    String[] files = dir.list();

    ArrayAdapter<String> adapter =
        new ArrayAdapter<>(
            this,
            android.R.layout.simple_list_item_1,
            files);

    lview.setAdapter(adapter);
}
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <application
        android:allowBackup="true"
        android:label="ListViewApp"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar">

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>

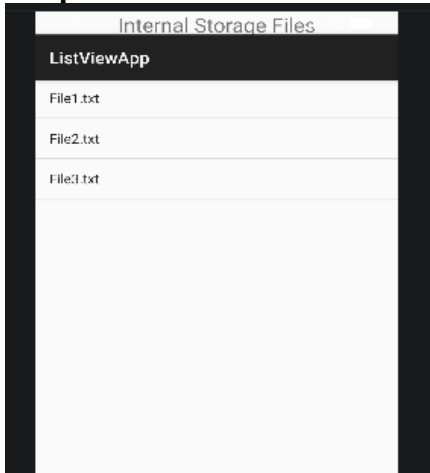
                <category
                    android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>
    </application>
</manifest>
```

</activity>

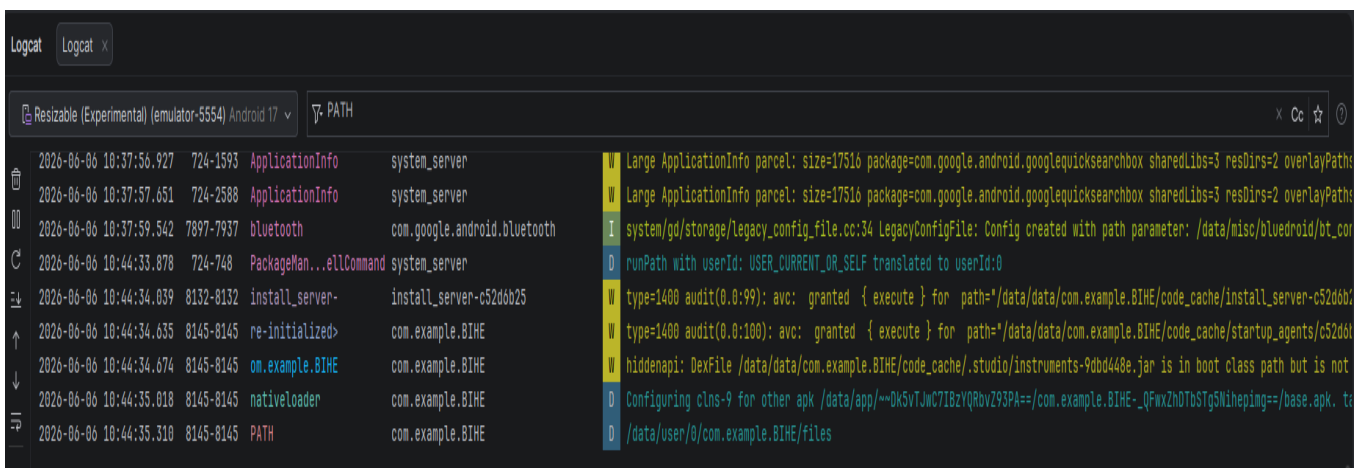
</application>

</manifest>

Output:



We can display the path where files are stored through **Logcat** option, by searching **PATH** keyword



6. Create an android application to display internal storage data in vertical format by using Custom Adapter.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <ListView
        android:id="@+id/lview"
        android:layout_width="match_parent"
        android:layout_height="match_parent"/>

</LinearLayout>
```

indiview.xml

(right click on **layout** then select **Layout Resource File**, name it as **indiview** and, root name as **LinearLayout** and save it)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:padding="10dp"
    android:orientation="horizontal">

    <ImageView
        android:id="@+id/iview"
        android:layout_width="60dp"
        android:layout_height="60dp"
        android:src="@mipmap/ic_launcher"/>

    <LinearLayout
        android:layout_width="0dp"
        android:layout_weight="1"
        android:layout_height="wrap_content"
        android:orientation="vertical">

        <TextView
            android:id="@+id/tv1"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="File Name"
            android:textSize="18sp"/>

    </LinearLayout>

</LinearLayout>
```

```

<TextView
    android:id="@+id/tv2"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="File Size"/>
</LinearLayout>

<Button
    android:id="@+id/b1"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Delete"/>

</LinearLayout>

```

MainActivity.java

```

package com.example.BIHE;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.widget.ListView;

import java.io.File;

public class MainActivity extends AppCompatActivity {

    ListView lview;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        lview = findViewById(R.id.lview);

        try {
            openFileOutput("File1.txt", MODE_PRIVATE).close();
            openFileOutput("File2.txt", MODE_PRIVATE).close();
            openFileOutput("File3.txt", MODE_PRIVATE).close();
        }
        catch(Exception e)
        {
            e.printStackTrace();
        }
    }
}

```

```

        File dir = getFilesDir();
        File[] files = dir.listFiles();

        MyAdapter adapter = new MyAdapter(this, files);
        lv.setAdapter(adapter);
    }
}

```

MyAdapter.java(create new java class file, name it as MyAdapter and save it)

```

package com.example.BIHE;

import android.app.Activity;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.BaseAdapter;
import android.widget.Button;
import android.widget.ImageView;
import android.widget.TextView;

import java.io.File;

public class MyAdapter extends BaseAdapter {

    Activity activity;
    File[] files;

    public MyAdapter(Activity activity, File[] files) {
        this.activity = activity;
        this.files = files;
    }

    @Override
    public int getCount() {
        return files.length;
    }

    @Override
    public Object getItem(int position) {
        return files[position];
    }

    @Override
    public long getItemId(int position) {
        return position;
    }
}

```

```

@Override
public View getView(final int position,
                    View convertView,
                    ViewGroup parent) {

    LayoutInflater inflater =
        activity.getLayoutInflater();

    View v =
        inflater.inflate(
            R.layout.indiview,
            null);

    ImageView iv =
        v.findViewById(R.id.iview);

    TextView tv1 =
        v.findViewById(R.id.tv1);

    TextView tv2 =
        v.findViewById(R.id.tv2);

    Button b1 =
        v.findViewById(R.id.b1);

    tv1.setText(files[position].getName());

    tv2.setText(
        files[position].length()
        + " bytes");

    b1.setOnClickListener(view -> {

        files[position].delete();

        File dir =
            activity.getFilesDir();

        files = dir.listFiles();

        notifyDataSetChanged();
    });

    return v;
}
}

```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android=
    "http://schemas.android.com/apk/res/android">

    <application
        android:allowBackup="true"
        android:label="CustomAdapterApp"
        android:theme="@style/Theme.AppCompat.Light">

        <activity
            android:name=".MainActivity"
            android:exported="true">

                <intent-filter>
                    <action
                        android:name=
                            "android.intent.action.MAIN"/>

                    <category
                        android:name=
                            "android.intent.category.LAUNCHER"/>
                </intent-filter>

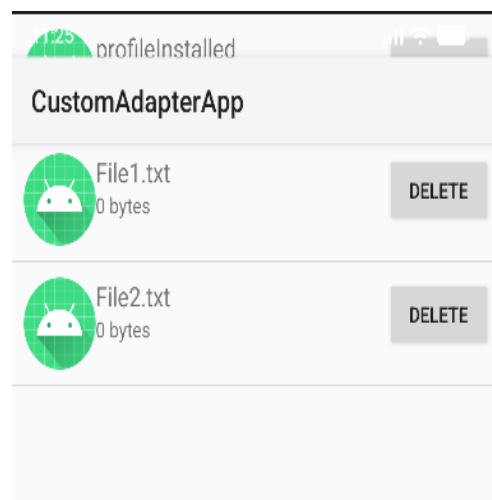
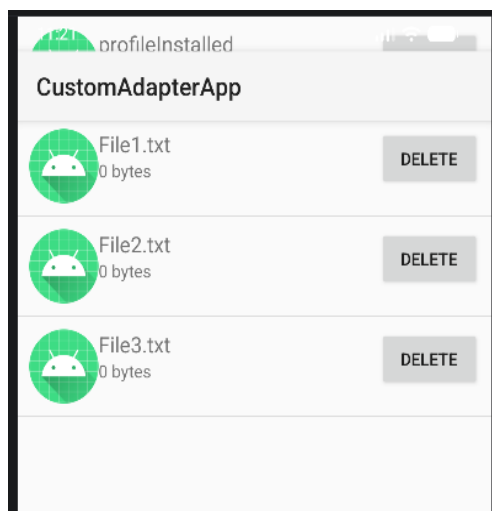
            </activity>

        </application>

    </manifest>
```

Output:

Click on DELETE Button to delete the file, it will automatically delete file at run time
For example: we deleted File3.txt file then it displays only first Two files.



7. Create an android application to display WhatsApp videos in grid view by using Custom Adapter.

Step1: create below path inside Device Explorer folder

/storage/emulated/0/Android/media/com.whatsapp/WhatsApp/Media/WhatsApp Video/



Step2: copy .mp4 vobex file inside **WhatsApp Video**

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<androidx.recyclerview.widget.RecyclerView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/recyclerView"
    android:layout_width="match_parent"
    android:layout_height="match_parent"/>
  
```

item_video.xml

```

<?xml version="1.0" encoding="utf-8"?>
<ImageView xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/imgVideo"
    android:layout_width="150dp"
    android:layout_height="150dp"
    android:layout_margin="100dp"
    android:scaleType="centerCrop"/>
  
```

MainActivity.java

```

package com.example.BIHE;

import android.Manifest;
import android.content.pm.PackageManager;
import android.os.Build;
import android.os.Bundle;

import androidx.annotation.NonNull;
  
```

```
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
import androidx.recyclerview.widget.GridLayoutManager;
import androidx.recyclerview.widget.RecyclerView;

import java.io.File;
import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    RecyclerView recyclerView;
    ArrayList<File> videoList = new ArrayList<>();

    private static final int REQUEST_CODE = 111;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        recyclerView = findViewById(R.id.recyclerView);
        recyclerView.setLayoutManager(new GridLayoutManager(this, 2));

        checkPermission();
    }

    private void checkPermission() {

        String permission;

        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
            permission = Manifest.permission.READ_MEDIA_VIDEO;
        }
    }
}
```

```
    } else {  
        permission = Manifest.permission.READ_EXTERNAL_STORAGE;  
    }  
  
    if (ContextCompat.checkSelfPermission(this, permission)  
        == PackageManager.PERMISSION_GRANTED) {  
  
        readFiles();  
  
    } else {  
  
        ActivityCompat.requestPermissions(  
            this,  
            new String[]{permission},  
            REQUEST_CODE);  
    }  
}  
  
@Override  
public void onRequestPermissionsResult(  
    int requestCode,  
    @NonNull String[] permissions,  
    @NonNull int[] grantResults) {  
  
    super.onRequestPermissionsResult(  
        requestCode,  
        permissions,  
        grantResults);  
  
    if (requestCode == REQUEST_CODE  
        && grantResults.length > 0  
        && grantResults[0] == PackageManager.PERMISSION_GRANTED) {
```

```
        readFiles();
    }
}

private void readFiles() {

    String path =
        "/storage/emulated/0/Android/media/com.whatsapp/WhatsApp/Media/WhatsApp Video/";

    File folder = new File(path);

    if (!folder.exists())
        return;

    File[] files = folder.listFiles();

    if (files == null)
        return;

    videoList.clear();

    for (File file : files) {

        if (file.isFile()
            && (file.getName().endsWith(".mp4")
                || file.getName().endsWith(".3gp"))) {

            videoList.add(file);
        }
    }

    recyclerView.setAdapter(
        new VideoAdapter(this, videoList));
}
```

```
}  
}
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>  
<manifest xmlns:android="http://schemas.android.com/apk/res/android">  
  
    <uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE" />  
    <uses-permission android:name="android.permission.READ_MEDIA_VIDEO" />  
  
    <application  
        android:allowBackup="true"  
        android:label="WhatsApp Videos"  
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar">  
  
        <activity  
            android:name=".MainActivity"  
            android:exported="true">  
  
            <intent-filter>  
                <action android:name="android.intent.action.MAIN" />  
                <category android:name="android.intent.category.LAUNCHER" />  
            </intent-filter>  
  
        </activity>  
    </application>  
  
</manifest>
```

VideoAdapter.java

```
package com.example.BIHE;  
  
import android.content.Context;  
import android.graphics.Bitmap;
```

```
import android.media.ThumbnailUtils;
import android.provider.MediaStore;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ImageView;

import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

import java.io.File;
import java.util.ArrayList;

public class VideoAdapter
    extends RecyclerView.Adapter<VideoAdapter.ViewHolder> {

    Context context;
    ArrayList<File> files;

    public VideoAdapter(Context context,
        ArrayList<File> files) {

        this.context = context;
        this.files = files;
    }

    @NonNull
    @Override
    public ViewHolder onCreateViewHolder(
        @NonNull ViewGroup parent,
        int viewType) {

        View view = LayoutInflater.from(context)
```

```

        .inflate(
            R.layout.item_video,
            parent,
            false);

    return new ViewHolder(view);
}

@Override
public void onBindViewHolder(
    @NonNull ViewHolder holder,
    int position) {

    File file = files.get(position);

    Bitmap bitmap = ThumbnailUtils.createVideoThumbnail(
        file.getAbsolutePath(),
        MediaStore.Images.Thumbnails.MINI_KIND);

    holder.imgVideo.setImageBitmap(bitmap);
}

@Override
public int getItemCount() {
    return files.size();
}

static class ViewHolder extends RecyclerView.ViewHolder {

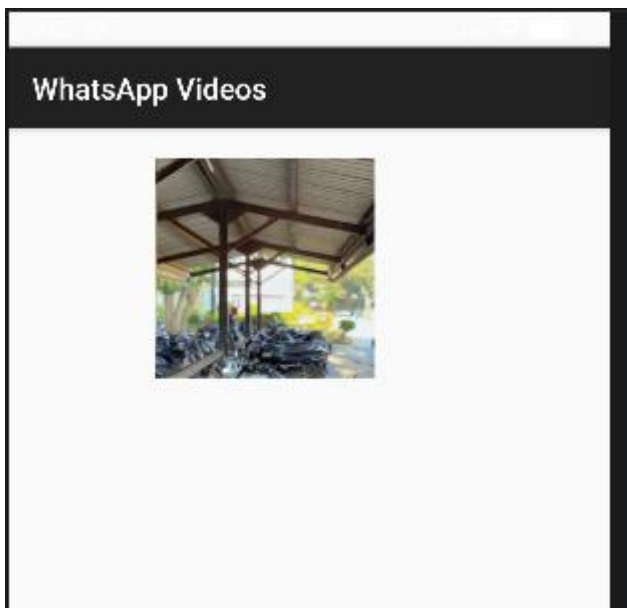
    ImageView imgVideo;

    public ViewHolder(@NonNull View itemView) {
        super(itemView);
    }
}

```

```
imgVideo = itemView.findViewById(R.id.imgVideo);  
}  
}  
}
```

Output:



8. Create an android application to display webpage by using Web view Component.**activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<WebView xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/webView"
    android:layout_width="match_parent"
    android:layout_height="match_parent" />
```

MainActivity.java

```
package com.example.BIHE;

import android.os.Bundle;
import android.webkit.WebView;
import android.webkit.WebViewClient;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    WebView webView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        webView = findViewById(R.id.webView);

        webView.setWebViewClient(new WebViewClient());

        webView.getSettings().setJavaScriptEnabled(true);
```

```
        webView.loadUrl("https://www.google.com");
    }
}
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>

<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <uses-permission android:name="android.permission.INTERNET"/>

    <application
        android:allowBackup="true"
        android:label="WebViewApp"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar">

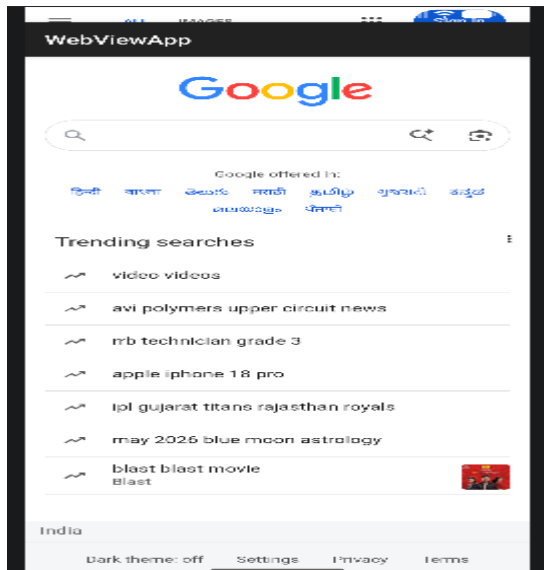
        <activity android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>

                <category
                    android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>

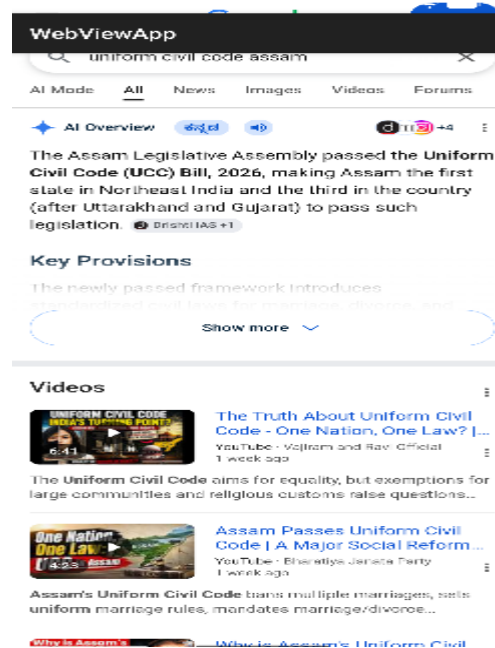
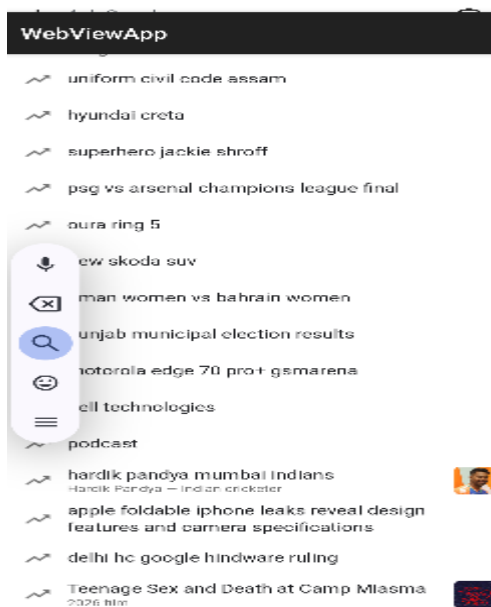
    </application>

</manifest>
```

Output:



Enter URL or choose from the list shown above



9. Create an android application to display different webpages in fragments by using Fragments**Component.****CourseFragment.java**

```
package com.example.BIHE;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;

import androidx.fragment.app.Fragment;

public class CourseFragment extends Fragment {

    @Override
    public View onCreateView(LayoutInflater inflater,
        ViewGroup container,
        Bundle savedInstanceState) {

        return inflater.inflate(
            R.layout.fragment_courses,
            container,
            false);
    }
}
```

FacultyFragment.java

```
package com.example.BIHE;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;

import androidx.fragment.app.Fragment;

public class FacultyFragment extends Fragment {

    @Override
    public View onCreateView(LayoutInflater inflater,
        ViewGroup container,
        Bundle savedInstanceState) {

        return inflater.inflate(
            R.layout.fragment_faculty,
            container,
            false);
    }
}
```

HomeFragment.java

```
package com.example.BIHE;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;

import androidx.fragment.app.Fragment;

public class HomeFragment extends Fragment {

    @Override
    public View onCreateView(LayoutInflater inflater,
                             ViewGroup container,
                             Bundle savedInstanceState) {

        return inflater.inflate(
            R.layout.fragment_home,
            container,
            false);
    }
}
```

MainActivity.java

```
package com.example.BIHE;

import android.os.Bundle;
import android.view.View;

import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.Fragment;
import androidx.fragment.app.FragmentManager;

public class MainActivity extends AppCompatActivity {

    FragmentManager fm;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        fm = getSupportFragmentManager();

        fm.beginTransaction()
            .replace(R.id.fragment, new HomeFragment())
    }
}
```

```

        .commit();
    }

    public void home(View v) {
        loadFragment(new HomeFragment());
    }

    public void courses(View v) {
        loadFragment(new CourseFragment());
    }

    public void faculties(View v) {
        loadFragment(new FacultyFragment());
    }

    private void loadFragment(Fragment fragment) {
        fm.beginTransaction()
            .replace(R.id.frag1, fragment)
            .commit();
    }
}

```

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">

        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Home"
            android:onClick="home"/>

        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Courses"
            android:onClick="courses"/>

        <Button

```

```

        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Faculty"
        android:onClick="faculties"/>
    </LinearLayout>

    <FrameLayout
        android:id="@+id/frag1"
        android:layout_width="match_parent"
        android:layout_height="match_parent"/>
</LinearLayout>

```

Fragment_course.xml

```

<?xml version="1.0" encoding="utf-8"?>
<TextView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:text="Courses Fragment"
    android:textSize="24sp"
    android:gravity="center"/>

```

Fragment_faculty.xml

```

<?xml version="1.0" encoding="utf-8"?>
<TextView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:text="Faculty Fragment"
    android:textSize="24sp"
    android:gravity="center"/>

```

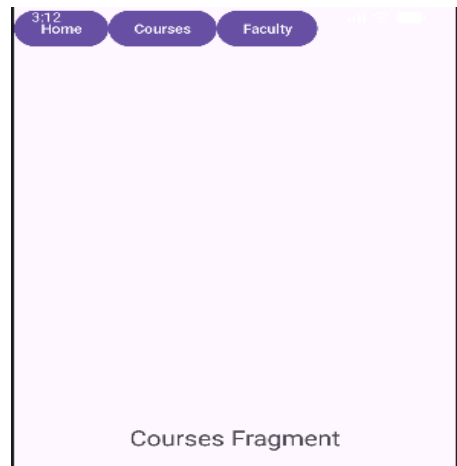
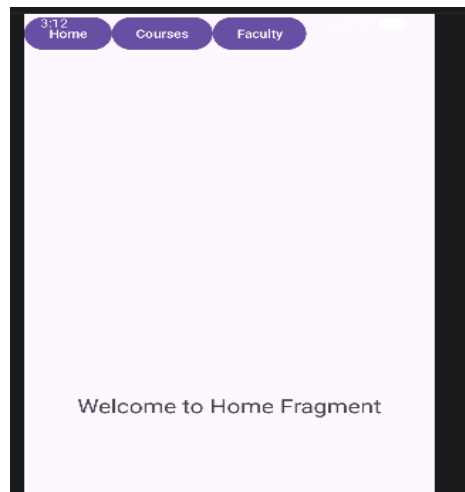
Fragment_home.xml

```

<?xml version="1.0" encoding="utf-8"?>
<TextView
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:text="Welcome to Home Fragment"
    android:textSize="24sp"
    android:gravity="center"/>

```

Output:



10.Create an android application to store data by using Shared Preferences.**activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/flayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent" />
```

fragment_login.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/etUser"
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:hint="Enter Username"/>

    <EditText
        android:id="@+id/etPass"
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:hint="Enter Password"
        android:inputType="textPassword"/>

    <Button
        android:id="@+id/btnSave"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Save"/>

    <TextView
        android:id="@+id/tvResult"
        android:layout_width="wrap_content"
        android:layout_height="48dp"
        android:textSize="18sp"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.BIHE;

import android.app.FragmentManager;
import android.app.FragmentTransaction;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        FragmentManager fm = getFragmentManager();
        FragmentTransaction tx = fm.beginTransaction();

        tx.add(R.id.flayout, new LoginFragment());
        tx.commit();
    }
}
```

LoginFragment.java

```
package com.example.BIHE;

import android.app.Fragment;
import android.content.SharedPreferences;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

public class LoginFragment extends Fragment {

    EditText etUser, etPass;
    Button btnSave;
    TextView tvResult;

    SharedPreferences sp;

    @Override
    public View onCreateView(LayoutInflater inflater,
                             ViewGroup container,
```

```

        Bundle savedInstanceState) {

    View v = inflater.inflate(
        R.layout.fragment_login,
        container,
        false);

    etUser = v.findViewById(R.id.etUser);
    etPass = v.findViewById(R.id.etPass);
    btnSave = v.findViewById(R.id.btnSave);
    tvResult = v.findViewById(R.id.tvResult);

    sp = getActivity().getSharedPreferences(
        "MyData",
        getActivity().MODE_PRIVATE);

    btnSave.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {

            String user = etUser.getText().toString();
            String pass = etPass.getText().toString();

            SharedPreferences.Editor editor = sp.edit();

            editor.putString("username", user);
            editor.putString("password", pass);

            editor.commit();

            tvResult.setText("Data Saved Successfully");
        }
    });

    return v;
}
}

```

AndroidManifest.xml

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.BIHE">

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"

```

```

android:theme="@style/Theme.AppCompat.Light.DarkActionBar">

<activity android:name=".MainActivity"
    android:exported="true">

    <intent-filter>
        <action android:name="android.intent.action.MAIN" />

        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>

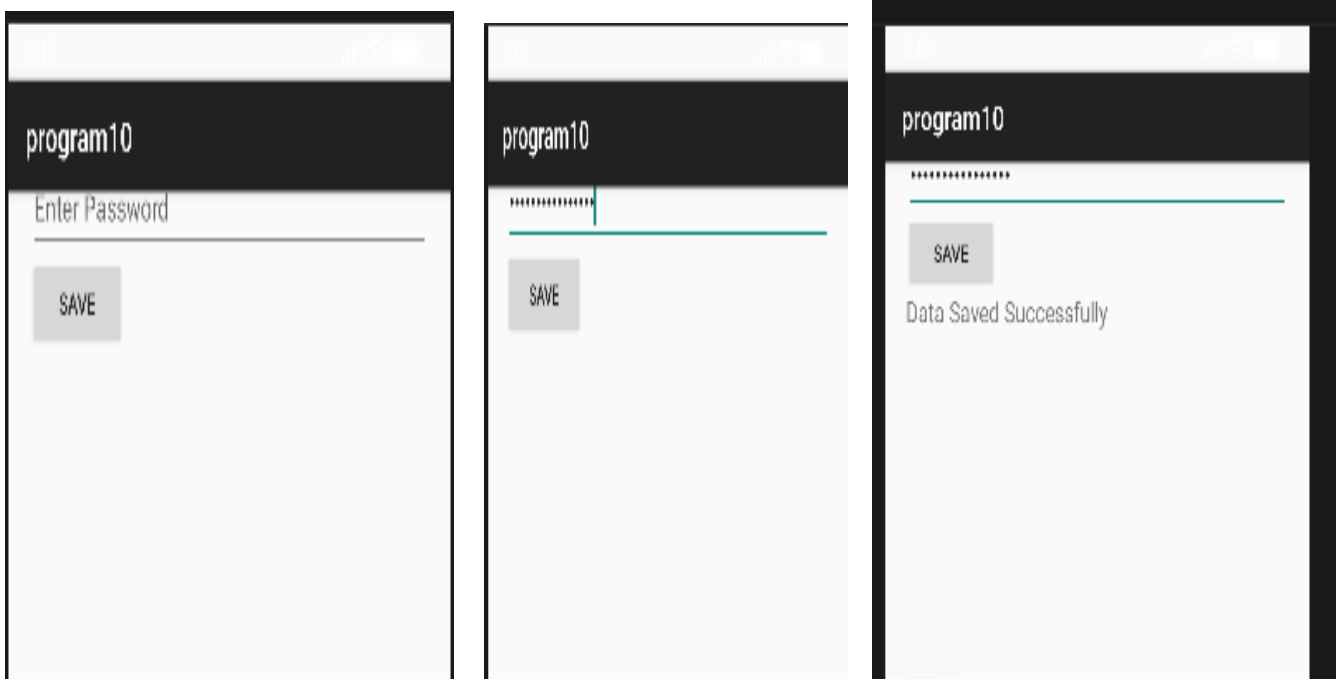
</activity>

</application>

</manifest>

```

Output:



11. Create an android application to demonstrate concept of SQLite Database storage method**MainActivity.java**

```
package com.example.BIHE;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.os.Bundle;
import android.view.View;
import android.widget.ArrayAdapter;
import android.widget.EditText;
import android.widget.ListView;
import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;

import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    EditText et1, et2, et3, et4;
    SQLiteDatabase dBase;
    ListView lview;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        et1 = findViewById(R.id.et1);
        et2 = findViewById(R.id.et2);
        et3 = findViewById(R.id.et3);
        et4 = findViewById(R.id.et4);

        lview = findViewById(R.id.lview);

        dBase = openOrCreateDatabase(
            "empdb",
            Context.MODE_PRIVATE,
            null);

        dBase.execSQL(
            "CREATE TABLE IF NOT EXISTS ttemp(" +
            "empid INTEGER PRIMARY KEY," +
            "empname TEXT," +
```

```

        "empdesig TEXT," +
        "empdept TEXT");
    }

    public void insert(View v) {

        ContentValues cv = new ContentValues();

        cv.put("empid",
            Integer.parseInt(et1.getText().toString()));
        cv.put("empname",
            et2.getText().toString());
        cv.put("empdesig",
            et3.getText().toString());
        cv.put("empdept",
            et4.getText().toString());

        long status = dBase.insert(
            "tbemp",
            null,
            cv);

        if(status != -1) {
            Toast.makeText(
                this,
                "Data Inserted",
                Toast.LENGTH_SHORT).show();

            read(v);
        } else {
            Toast.makeText(
                this,
                "Insert Failed",
                Toast.LENGTH_SHORT).show();
        }
    }

    public void read(View v) {

        Cursor c = dBase.query(
            "tbemp",
            null,
            null,
            null,
            null,
            null,
            null);

        ArrayList<String> list =

```

```

        new ArrayList<>();

while(c.moveToNext()) {

    String msg =
        c.getInt(0) + " | " +
        c.getString(1) + " | " +
        c.getString(2) + " | " +
        c.getString(3);

    list.add(msg);
}

c.close();

ArrayAdapter<String> adapter =
    new ArrayAdapter<>(
        this,
        android.R.layout.simple_list_item_1,
        list);

lview.setAdapter(adapter);
}

public void update(View v) {

    ContentValues cv = new ContentValues();

    cv.put("empname",
        et2.getText().toString());
    cv.put("empdesig",
        et3.getText().toString());
    cv.put("empdept",
        et4.getText().toString());

    int status = dBase.update(
        "tbemp",
        cv,
        "empid=?",
        new String[] {
            et1.getText().toString()
        });

    if(status > 0) {
        Toast.makeText(
            this,
            "Updated Successfully",
            Toast.LENGTH_SHORT).show();
    }
}

```

```

        read(v);
    } else {
        Toast.makeText(
            this,
            "Update Failed",
            Toast.LENGTH_SHORT).show();
    }
}

public void delete(View v) {

    int status = dBase.delete(
        "tbemp",
        "empid=?",
        new String[] {
            et1.getText().toString()
        });

    if(status > 0) {
        Toast.makeText(
            this,
            "Deleted Successfully",
            Toast.LENGTH_SHORT).show();

        read(v);
    } else {
        Toast.makeText(
            this,
            "Delete Failed",
            Toast.LENGTH_SHORT).show();
    }
}

}

```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="10dp">

    <EditText
        android:id="@+id/et1"
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:hint="Employee ID"
        android:inputType="number"/>

    <EditText
        android:id="@+id/et2"
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:hint="Employee Name"/>

    <EditText
        android:id="@+id/et3"
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:hint="Employee Designation"/>

    <EditText
        android:id="@+id/et4"
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:hint="Employee Department"/>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">

        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Insert"
            android:onClick="insert"/>

        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Read"
```

```
android:onClick="read"/>
```

```
<Button
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Update"
    android:onClick="update"/>
```

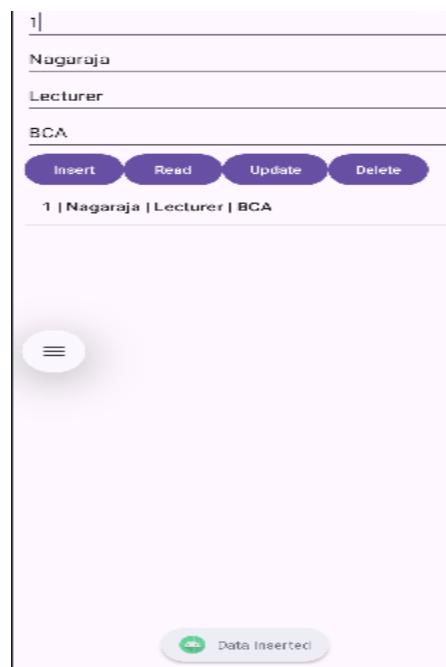
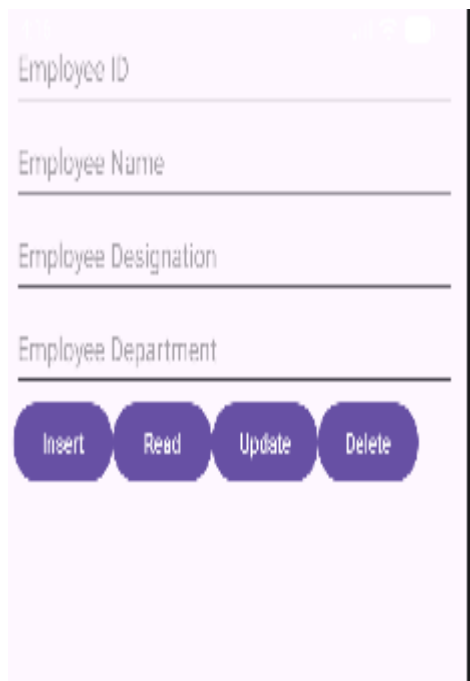
```
<Button
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Delete"
    android:onClick="delete"/>
```

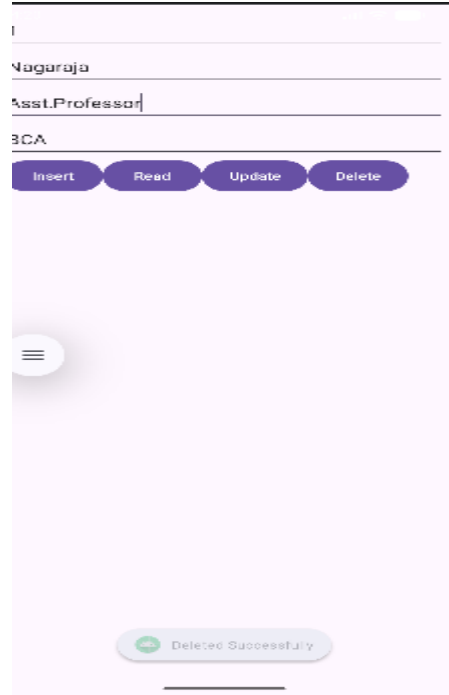
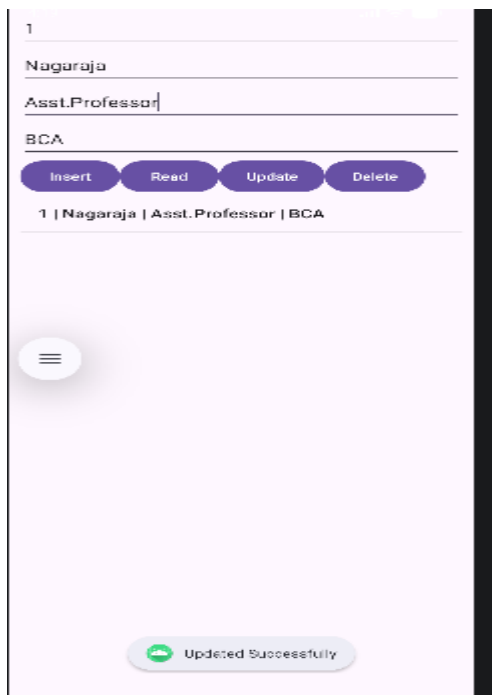
```
</LinearLayout>
```

```
<ListView
    android:id="@+id/lview"
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1"/>
```

```
</LinearLayout>
```

Output:





12. Write an android program to develop Video view application

Step 1: Create raw Folder

Create:

```
app
├── src
│   └── main
│       └── res
│           └── raw
│               └── sample.mp4
```

Copy any MP4 video and rename it sample.mp4.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">
```

```
    <VideoView
        android:id="@+id/videoView"
        android:layout_width="match_parent"
        android:layout_height="300dp" />
```

```
</LinearLayout>
```

MainActivity.java

```
package com.example.BIHE;

import android.net.Uri;
import android.os.Bundle;
import android.widget.MediaController;
import android.widget.VideoView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    VideoView videoView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        videoView = findViewById(R.id.videoView);

        String path =
            "android.resource://" +
                getPackageName() +
                "/" +
                R.raw.sample;

        Uri uri = Uri.parse(path);

        videoView.setVideoURI(uri);

        MediaController mediaController =
            new MediaController(this);

        mediaController.setAnchorView(videoView);

        videoView.setMediaController(
            mediaController);

        videoView.start();
    }
}
```

Output:



PART B

- 1. Create an android application to perform different types of operations (Send SMS, Making call and sending email) by using Telephony app.**

activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <EditText
        android:id="@+id/etPhone"
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:hint="Phone Number" />

    <EditText
        android:id="@+id/etMessage"
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:hint="Message" />

    <EditText
        android:id="@+id/etEmail"
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:hint="Email Address" />

    <Button
        android:id="@+id/btnSms"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Send SMS" />

    <Button
        android:id="@+id/btnCall"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Make Call" />

    <Button
        android:id="@+id/btnEmail"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Send Email" />

</LinearLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <application
        android:allowBackup="true"
        android:label="TelephonyApp"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar">

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>

        </activity>

    </application>

</manifest>
```

MainActivity.java

```
package com.example.BIHE;

import static com.example.BIHE.R.*;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText etPhone, etMessage, etEmail;
    Button btnSms, btnCall, btnEmail;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etPhone = findViewById(R.id.etPhone);
```

```

etMessage = findViewById(R.id.etMessage);
etEmail = findViewById(R.id.etEmail);
btnSms = findViewById(R.id.btnSms);
btnCall = findViewById(R.id.btnCall);
btnEmail = findViewById(R.id.btnEmail);

btnSms.setOnClickListener(v -> {

    Intent intent = new Intent(Intent.ACTION_VIEW);
    intent.setData(Uri.parse("sms:" + etPhone.getText().toString()));
    intent.putExtra("sms_body", etMessage.getText().toString());

    startActivity(intent);
});

btnCall.setOnClickListener(v -> {

    Intent intent = new Intent(
        Intent.ACTION_DIAL,
        Uri.parse("tel:" + etPhone.getText().toString()));

    startActivity(intent);
});

btnEmail.setOnClickListener(v -> {

    Intent intent = new Intent(Intent.ACTION_SEND);

    intent.setType("message/rfc822");

    intent.putExtra(
        Intent.EXTRA_EMAIL,
        new String[]{etEmail.getText().toString()});

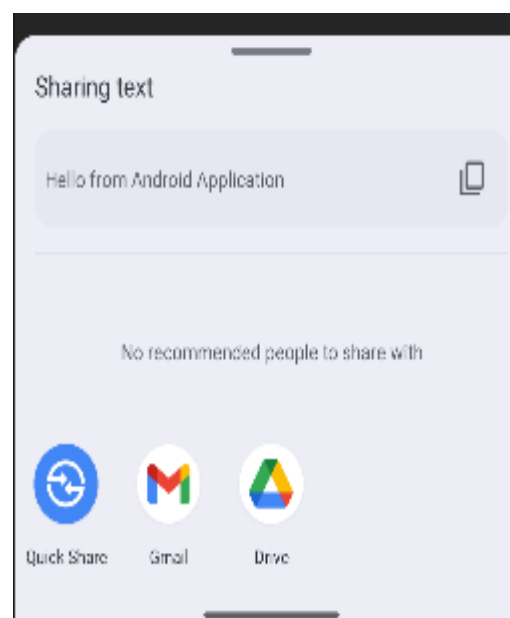
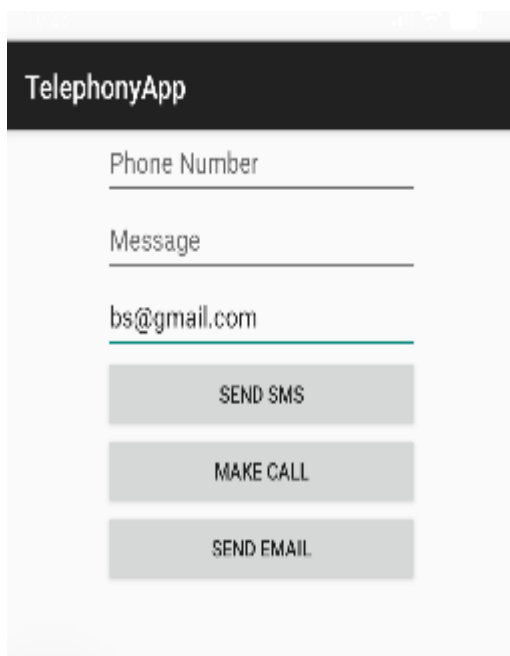
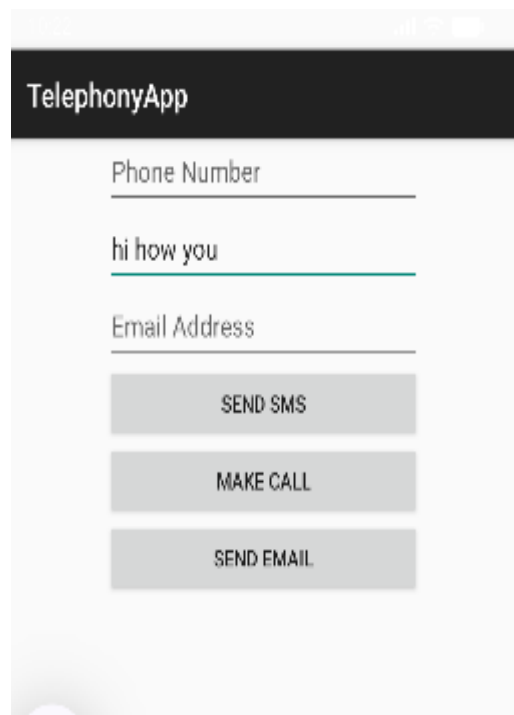
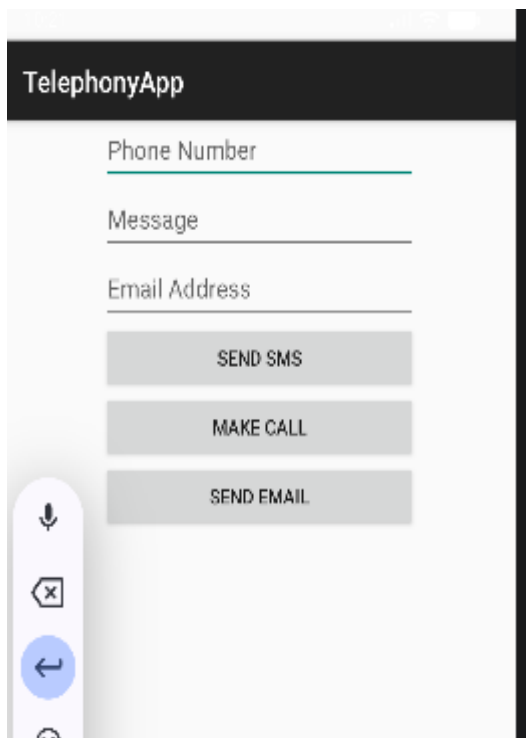
    intent.putExtra(
        Intent.EXTRA_SUBJECT,
        "Test Mail");

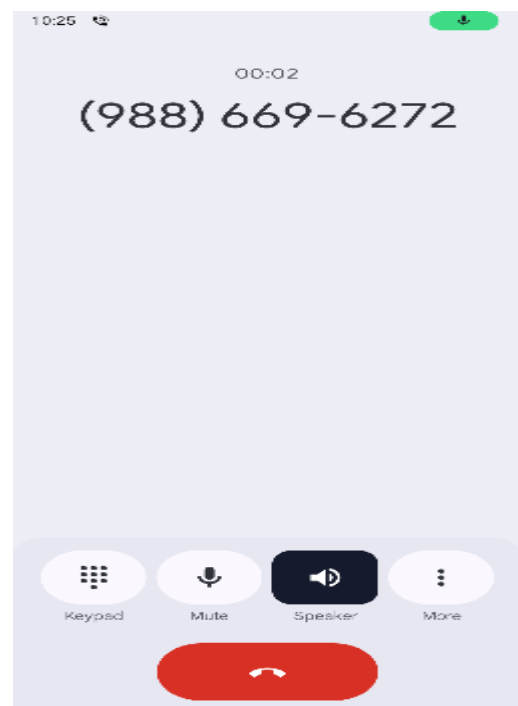
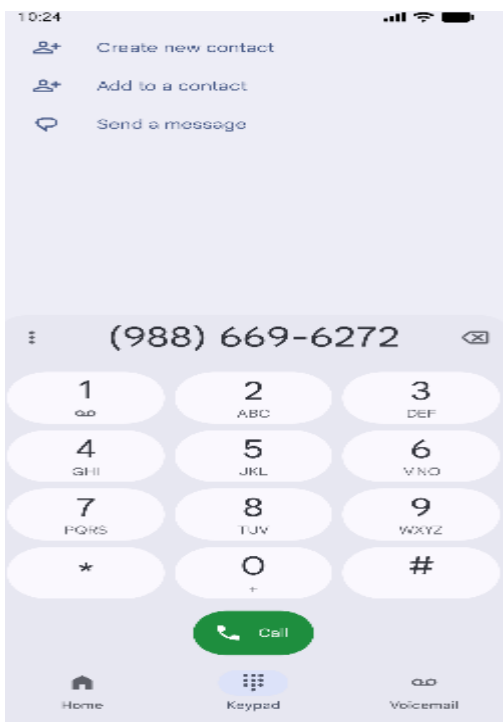
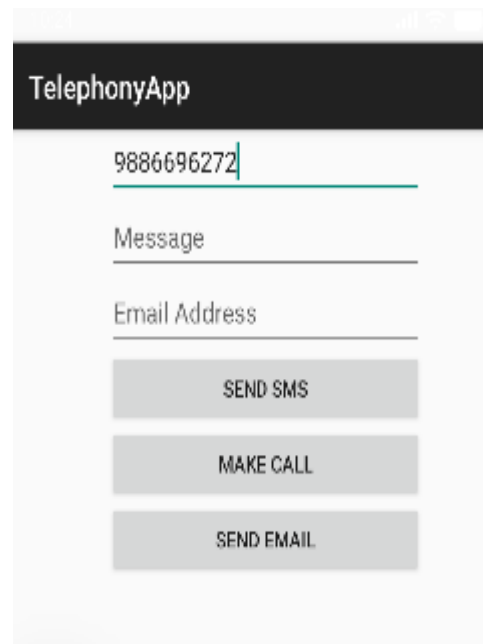
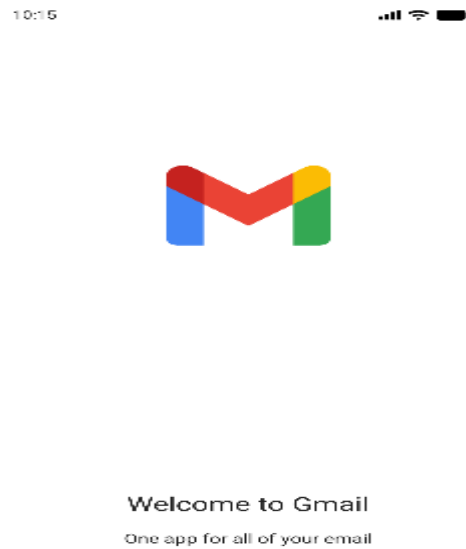
    intent.putExtra(
        Intent.EXTRA_TEXT,
        "Hello from Android Application");

    startActivity(
        Intent.createChooser(
            intent,
            "Select Email App"));
});
}
}

```

Output:





2. Write an android program to develop Media player application.

Step 1:

app/

└─ java/

 └─ com.example.BIHE/

 └─ MainActivity.java

└─ res/

 └─ layout/activity_main.xml

 └─ raw/song.mp3 (IMPORTANT)

Step2:

Create folder:

```
res → raw
```

Then add any file:

```
song.mp3
```

(You can copy any small mp3 file)

Step3:

Place your image here:

```
res → drawable → bg_music.jpg
```

(Use any image, rename it to lowercase like `bg_music.jpg`)

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <!-- Background Image -->
    <ImageView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:src="@drawable/bg_music"
        android:scaleType="centerCrop" />

    <!-- Foreground Controls -->
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:gravity="center"
        android:padding="20dp"
        android:background="#80000000">

        <Button
            android:id="@+id/btnPlay"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Play"
            android:textColor="#FFFFFF"/>

        <Button
            android:id="@+id/btnPause"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
```

```

        android:text="Pause"
        android:textColor="#FFFFFF"/>

```

```

<Button
    android:id="@+id/btnStop"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Stop"
    android:textColor="#FFFFFF"/>

```

```

</LinearLayout>

```

```

</FrameLayout>

```

MainActivity.java

```

package com.example.BIHE;

import android.media.MediaPlayer;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    MediaPlayer mediaPlayer;
    Button btnPlay, btnPause, btnStop;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

```

```
setContentView(R.layout.activity_main);

btnPlay = findViewById(R.id.btnPlay);
btnPause = findViewById(R.id.btnPause);
btnStop = findViewById(R.id.btnStop);

mediaPlayer = MediaPlayer.create(this, R.raw.song);

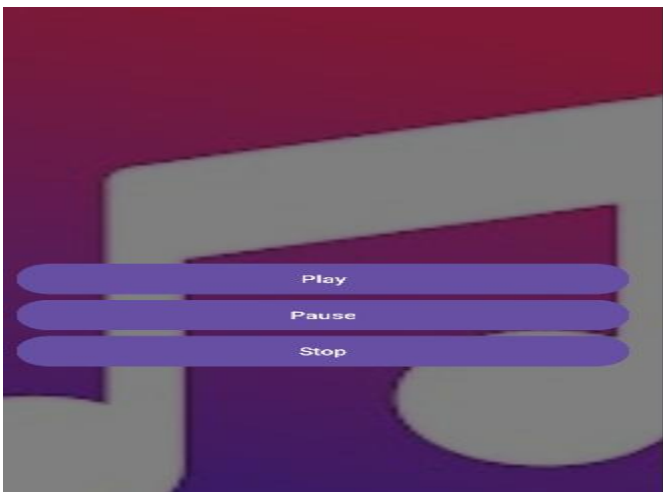
btnPlay.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        mediaPlayer.start();
    }
});

btnPause.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        if (mediaPlayer.isPlaying()) {
            mediaPlayer.pause();
        }
    }
});

btnStop.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {
        mediaPlayer.stop();
        mediaPlayer = MediaPlayer.create(MainActivity.this, R.raw.song);
    }
});
}
```

```
@Override  
protected void onDestroy() {  
    super.onDestroy();  
    if (mediaPlayer != null) {  
        mediaPlayer.release();  
    }  
}
```

Output:



3. Create an android application to display current location on Google maps by using Google-Maps Service.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<WebView xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/webView"
    android:layout_width="match_parent"
    android:layout_height="match_parent"/>
```

MainActivity.java

```
package com.example.BIHE;

import android.os.Bundle;
import android.webkit.WebView;
import android.webkit.WebViewClient;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    WebView webView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        webView = findViewById(R.id.webView);

        webView.setWebViewClient(new WebViewClient());

        String lat = "14.4644"; // example: Davangere latitude
        String lon = "75.9218"; // example: Davangere longitude
```

```
String url =
    "https://www.google.com/maps/@"
    + lat + "," + lon + ",15z";

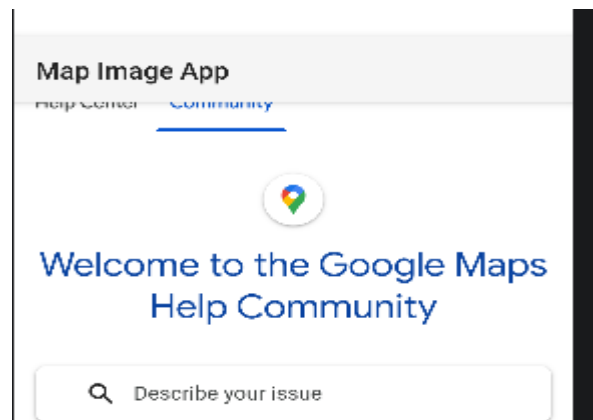
webView.loadUrl(url);
}
}
```

AndroidManifest.xml

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android">
    <uses-permission android:name="android.permission.INTERNET"/>
    <application
        android:allowBackup="true"
        android:label="Map Image App"
        android:theme="@style/Theme.AppCompat.Light">

        <activity android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
                <category android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>
        </activity>
    </application>
</manifest>
```

Output:



4. Write an android program to develop Audio Recording application

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <uses-permission android:name="android.permission.RECORD_AUDIO"/>

    <application
        android:allowBackup="true"
        android:label="AudioRecorder"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar">

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>

                <category
                    android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>

        </activity>

    </application>

</manifest>
```

MainActivity.java

```
package com.example.BIHE;

import android.Manifest;
import android.content.pm.PackageManager;
import android.media.MediaPlayer;
import android.media.MediaRecorder;
import android.os.Bundle;
import android.os.Environment;
import android.widget.Button;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;
```

```
import java.io.IOException;

public class MainActivity extends AppCompatActivity {

    Button btnRecord, btnStop, btnPlay;

    MediaRecorder mediaRecorder;
    MediaPlayer mediaPlayer;

    String fileName;

    private static final int REQUEST_PERMISSION_CODE = 1000;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnRecord = findViewById(R.id.btnRecord);
        btnStop = findViewById(R.id.btnStop);
        btnPlay = findViewById(R.id.btnPlay);

        fileName = getExternalFilesDir(null).getAbsolutePath()
            + "/recorded_audio.3gp";

        if (!checkPermissions()) {
            requestPermissions();
        }

        btnRecord.setOnClickListener(v -> startRecording());

        btnStop.setOnClickListener(v -> stopRecording());

        btnPlay.setOnClickListener(v -> playRecording());
    }

    private void startRecording() {

        mediaRecorder = new MediaRecorder();

        mediaRecorder.setAudioSource(MediaRecorder.AudioSource.MIC);
        mediaRecorder.setOutputFormat(MediaRecorder.OutputFormat.THREE_GPP);
        mediaRecorder.setAudioEncoder(MediaRecorder.AudioEncoder.AMR_NB);
        mediaRecorder.setOutputFile(fileName);

        try {
            mediaRecorder.prepare();
            mediaRecorder.start();
            Toast.makeText(this,
```

```

        "Recording Started",
        Toast.LENGTH_SHORT).show();

    } catch (IOException e) {
        e.printStackTrace();
    }
}

private void stopRecording() {

    if (mediaRecorder != null) {
        mediaRecorder.stop();
        mediaRecorder.release();
        mediaRecorder = null;

        Toast.makeText(this,
            "Recording Saved",
            Toast.LENGTH_SHORT).show();
    }
}

private void playRecording() {

    mediaPlayer = new MediaPlayer();

    try {
        mediaPlayer.setDataSource(fileName);
        mediaPlayer.prepare();
        mediaPlayer.start();

        Toast.makeText(this,
            "Playing Audio",
            Toast.LENGTH_SHORT).show();

    } catch (IOException e) {
        e.printStackTrace();
    }
}

private boolean checkPermissions() {
    return ContextCompat.checkSelfPermission(
        this,
        Manifest.permission.RECORD_AUDIO)
        == PackageManager.PERMISSION_GRANTED;
}

private void requestPermissions() {
    ActivityCompat.requestPermissions(
        this,

```

```

        new String[]{
            Manifest.permission.RECORD_AUDIO
        },
        REQUEST_PERMISSION_CODE);
    }

    @Override
    public void onRequestPermissionsResult(
        int requestCode,
        @NonNull String[] permissions,
        @NonNull int[] grantResults) {

        super.onRequestPermissionsResult(
            requestCode,
            permissions,
            grantResults);

        if (requestCode == REQUEST_PERMISSION_CODE) {

            if (grantResults.length > 0
                && grantResults[0]
                == PackageManager.PERMISSION_GRANTED) {

                Toast.makeText(this,
                    "Permission Granted",
                    Toast.LENGTH_SHORT).show();

            } else {

                Toast.makeText(this,
                    "Permission Denied",
                    Toast.LENGTH_SHORT).show();

            }
        }
    }
}

```

AndroidManifest.xml

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <uses-permission android:name="android.permission.RECORD_AUDIO"/>

    <application
        android:allowBackup="true"
        android:label="AudioRecorder"
        android:supportRtl="true"
        android:theme="@style/Theme.AppCompat.Light.DarkActionBar">

```

```

<activity
    android:name=".MainActivity"
    android:exported="true">

    <intent-filter>
        <action android:name="android.intent.action.MAIN"/>

        <category
            android:name="android.intent.category.LAUNCHER"/>
    </intent-filter>

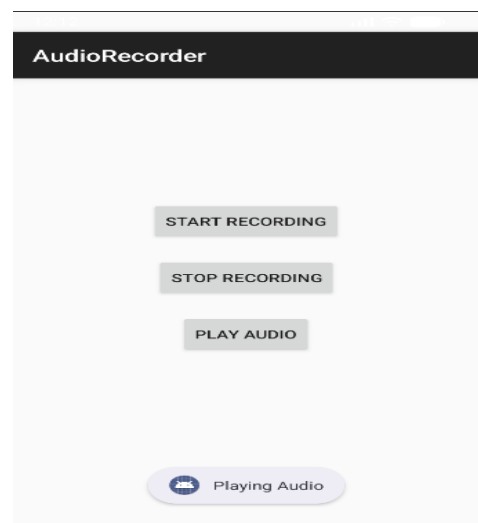
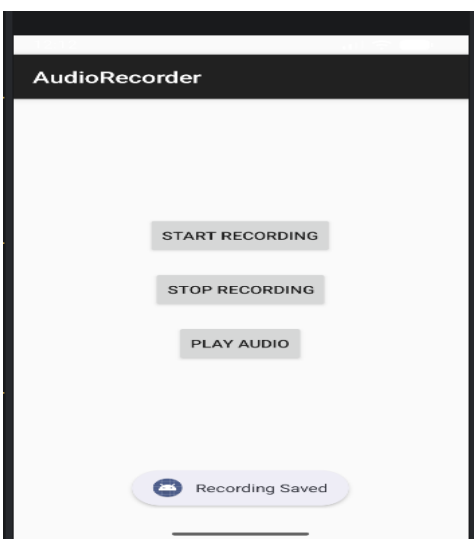
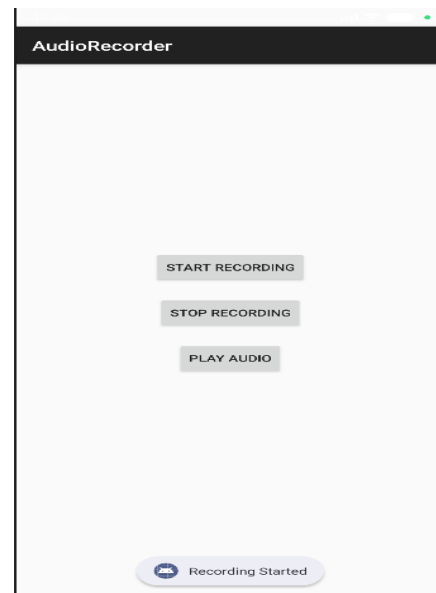
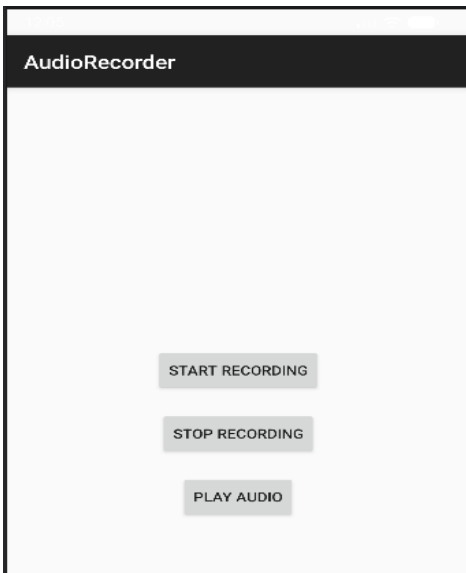
</activity>

</application>

</manifest>

```

Output:



5. Write an android program to develop Video Recording application.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <Button
        android:id="@+id/btnRecord"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Record Video"/>

    <VideoView
        android:id="@+id/videoView"
        android:layout_width="match_parent"
        android:layout_height="300dp"
        android:layout_marginTop="20dp"/>

</LinearLayout>
```

MainActivity.java

```
package com.example.BIHE;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.provider.MediaStore;
import android.widget.Button;
import android.widget.MediaController;
import android.widget.VideoView;

import androidx.activity.result.ActivityResultLauncher;
import androidx.activity.result.contract.ActivityResultContracts;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnRecord;
    VideoView videoView;

    ActivityResultLauncher<Intent> videoLauncher;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

```

btnRecord = findViewById(R.id.btnRecord);
videoView = findViewById(R.id.videoView);

videoLauncher = registerForActivityResult(
    new ActivityResultContracts.StartActivityForResult(),
    result -> {
        if (result.getResultCode() == RESULT_OK
            && result.getData() != null) {

            Uri videoUri = result.getData().getData();

            videoView.setVideoURI(videoUri);

            MediaController mediaController =
                new MediaController(this);

            mediaController.setAnchorView(videoView);

            videoView.setMediaController(mediaController);

            videoView.start();
        }
    });

btnRecord.setOnClickListener(v -> {

    Intent intent =
        new Intent(MediaStore.ACTION_VIDEO_CAPTURE);

    videoLauncher.launch(intent);
});
}
}

```

AndroidManifest.xml

```

<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <uses-permission android:name="android.permission.CAMERA" />

    <uses-feature
        android:name="android.hardware.camera"
        android:required="false" />

    <application
        android:allowBackup="true"
        android:label="VideoRecorder"
        android:theme="@style/Theme.AppCompat">

        <activity
            android:name=".MainActivity"
            android:exported="true">

```

```
<intent-filter>  
    <action android:name="android.intent.action.MAIN" />  
    <category android:name="android.intent.category.LAUNCHER" />  
</intent-filter>
```

```
</activity>
```

```
</application>
```

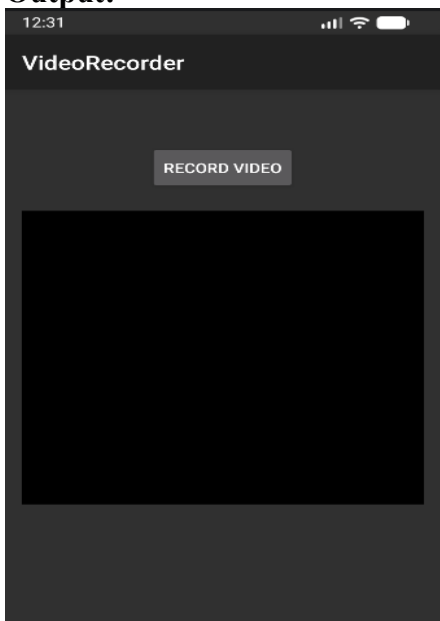
```
</manifest>
```

build.gradle (Module: app)

Ensure these dependencies exist:

```
dependencies {  
  
    implementation 'androidx.appcompat:appcompat:1.7.0'  
    implementation 'com.google.android.material:material:1.12.0'  
  
}
```

Output:



6. Write an android program to develop Camera and Gallery applications.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Camera"
        android:onClick="camera"/>

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Gallery"
        android:onClick="gallery"/>

    <ImageView
        android:id="@+id/ivew1"
        android:layout_width="250dp"
        android:layout_height="250dp" />

</LinearLayout>
```

MainActivity.java

```
package com.example.BIHE;

import android.content.Intent;
import android.graphics.Bitmap;
import android.net.Uri;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import android.view.View;
import android.widget.ImageView;

public class MainActivity extends AppCompatActivity {

    ImageView iview;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

```

        iview = findViewById(R.id.iview1);
    }

    public void camera(View v) {
        Intent i = new Intent("android.media.action.IMAGE_CAPTURE");
        startActivityForResult(i, 111);
    }

    public void gallery(View v) {
        Intent i = new Intent();
        i.setAction(Intent.ACTION_GET_CONTENT);
        i.setType("image/*");
        startActivityForResult(i, 123);
    }

    @Override
    protected void onActivityResult(int requestCode, int resultCode, Intent data) {
        super.onActivityResult(requestCode, resultCode, data);

        if (resultCode == RESULT_OK && data != null) {

            if (requestCode == 111) {
                Bitmap bmp = (Bitmap) data.getExtras().get("data");
                iview.setImageBitmap(bmp);
            }

            if (requestCode == 123) {
                Uri u = data.getData();
                iview.setImageURI(u);
            }
        }
    }
}

```

AndroidManifest.xml

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <uses-permission android:name="android.permission.CAMERA"/>

    <uses-feature
        android:name="android.hardware.camera"
        android:required="false"/>

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"

```

```

android:label="@string/app_name"
android:roundIcon="@mipmap/ic_launcher_round"
android:supportRtl="true"
android:theme="@style/Theme.Material3.DayNight.NoActionBar">

<activity
    android:name=".MainActivity"
    android:exported="true">

    <intent-filter>
        <action android:name="android.intent.action.MAIN"/>

        <category android:name="android.intent.category.LAUNCHER"/>
    </intent-filter>

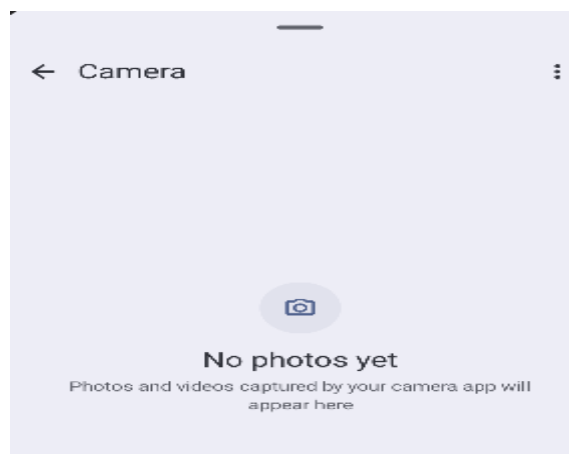
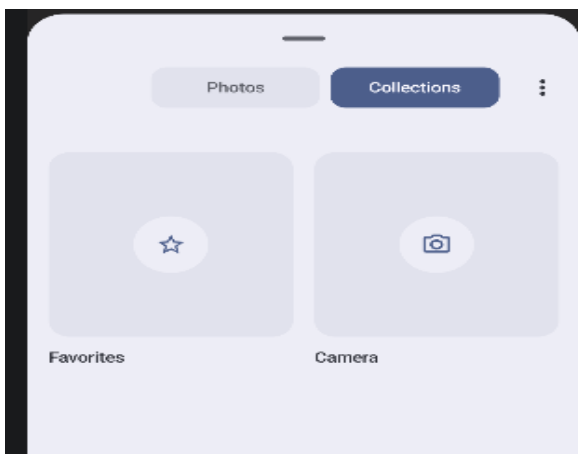
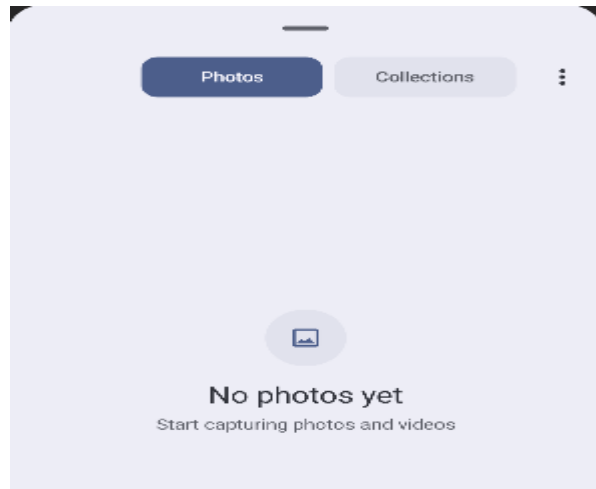
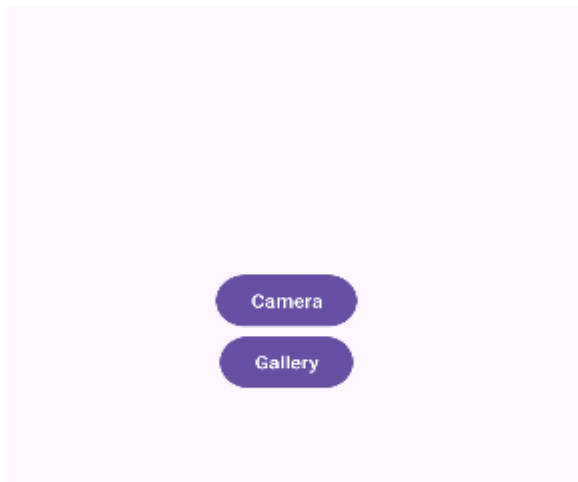
</activity>

</application>

</manifest>

```

Output:



7. Create an android application to get latitude and longitude value by using Location Service

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/tv1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Latitude"
        android:textSize="30sp"
        android:textStyle="bold"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="50dp"/>

    <TextView
        android:id="@+id/tv2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Longitude"
        android:textSize="30sp"
        android:textStyle="bold"
        android:layout_below="@id/tv1"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="50dp"/>

</RelativeLayout>
```

MainActivity.java

```
package com.example.BIHE;

import android.Manifest;
import android.content.Context;
import android.content.pm.PackageManager;
import android.location.Location;
import android.location.LocationListener;
import android.location.LocationManager;
import android.os.Bundle;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
```

```

TextView tv1, tv2;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    tv1 = findViewById(R.id.tv1);
    tv2 = findViewById(R.id.tv2);

    LocationManager lManager =
        (LocationManager) getSystemService(Context.LOCATION_SERVICE);

    if (checkSelfPermission(Manifest.permission.ACCESS_FINE_LOCATION)
        != PackageManager.PERMISSION_GRANTED &&
        checkSelfPermission(Manifest.permission.ACCESS_COARSE_LOCATION)
        != PackageManager.PERMISSION_GRANTED) {

        requestPermissions(
            new String[]{
                Manifest.permission.ACCESS_FINE_LOCATION,
                Manifest.permission.ACCESS_COARSE_LOCATION
            },
            100);
        return;
    }

    lManager.requestLocationUpdates(
        LocationManager.NETWORK_PROVIDER,
        1000,
        1,
        new LocationListener() {
            @Override
            public void onLocationChanged(Location location) {

                double lati = location.getLatitude();
                double longi = location.getLongitude();

                tv1.setText("Latitude : " + lati);
                tv2.setText("Longitude : " + longi);
            }
        });
}
}

```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"/>
    <uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"/>

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportRtl="true"
        android:theme="@style/Theme.Material3.DayNight.NoActionBar">

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>

                <category android:name="android.intent.category.LAUNCHER"/>
            </intent-filter>

        </activity>

    </application>

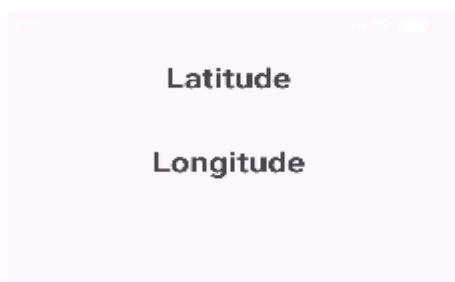
</manifest>
```

build.gradle.kts

```
fun implementation(string: String) {}

dependencies {
    implementation("com.google.android.gms:play-services-location:21.3.0")
}
```

Output:



8. Create an android application to get the Bluetooth devices and list of devices using Bluetooth and Vibrator Service.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <Switch
        android:id="@+id/s1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:text="BLUETOOTH"
        android:textSize="25sp" />

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:onClick="getBTDevices"
        android:text="GET BT DEVICES"
        android:textSize="25sp" />

    <Button
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_gravity="center"
        android:onClick="vibrate"
        android:text="VIBRATE"
        android:textSize="25sp" />
```

```
<ListView  
    android:id="@+id/lview"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent" />
```

```
</LinearLayout>
```

MainActivity.java

```
package com.example.BIHE;  
  
import android.bluetooth.BluetoothAdapter;  
import android.bluetooth.BluetoothDevice;  
import android.content.BroadcastReceiver;  
import android.content.Context;  
import android.content.Intent;  
import android.content.IntentFilter;  
import android.os.Bundle;  
import android.os.VibrationEffect;  
import android.os.Vibrator;  
import android.widget.AdapterView;  
import android.widget.CompoundButton;  
import android.widget.ListView;  
import android.widget.Switch;  
import android.annotation.SuppressLint;  
  
import androidx.appcompat.app.AppCompatActivity;  
  
import java.util.ArrayList;  
  
public class MainActivity extends AppCompatActivity {  
  
    Switch s1;
```

```

ListView lview;
BluetoothAdapter bAdapter;

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    s1 = findViewById(R.id.s1);
    lview = findViewById(R.id.lview);

    bAdapter = BluetoothAdapter.getDefaultAdapter();

    if (bAdapter != null) {
        s1.setChecked(bAdapter.isEnabled());

        s1.setOnCheckedChangeListener(
            new CompoundButton.OnCheckedChangeListener() {
                @Override
                public void onCheckedChanged(
                    CompoundButton buttonView,
                    boolean isChecked) {

                    if (isChecked) {
                        bAdapter.enable();
                    } else {
                        bAdapter.disable();
                    }
                }
            }
        );
    }
}

```

```

public void getBTDevices(android.view.View v) {

    final ArrayList<String> list = new ArrayList<>();

    final ArrayAdapter<String> adapter =
        new ArrayAdapter<>(
            this,
            android.R.layout.simple_list_item_1,
            list);

    lv.setAdapter(adapter);

    IntentFilter filter =
        new IntentFilter(BluetoothDevice.ACTION_FOUND);

    registerReceiver(new BroadcastReceiver() {
        @Override
        public void onReceive(
            Context context,
            Intent intent) {

            BluetoothDevice device =
                intent.getParcelableExtra(
                    BluetoothDevice.EXTRA_DEVICE);

            if (device != null) {

                String name = device.getName();
                if (name == null) {
                    name = "Unknown Device";
                }

                list.add(

```

```

        name + "\n" +
            device.getAddress());

        adapter.notifyDataSetChanged();
    }
}
}, filter);

if (bAdapter != null) {
    bAdapter.startDiscovery();
}
}

public void vibrate(android.view.View v) {

    Vibrator vib =
        (Vibrator) getSystemService(
            Context.VIBRATOR_SERVICE);

    if (vib != null) {

        if (android.os.Build.VERSION.SDK_INT >=
            android.os.Build.VERSION_CODES.O) {

            vib.vibrate(
                VibrationEffect.createOneShot(
                    1000,
                    VibrationEffect.DEFAULT_AMPLITUDE));

        } else {

            vib.vibrate(1000);
        }
    }
}

```

```

    }
}
}

```

AndroidManifest.xml

```

<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <!-- Bluetooth Permissions -->
    <uses-permission android:name="android.permission.BLUETOOTH"/>
    <uses-permission android:name="android.permission.BLUETOOTH_ADMIN"/>

    <!-- Android 12+ -->
    <uses-permission android:name="android.permission.BLUETOOTH_CONNECT"/>
    <uses-permission android:name="android.permission.BLUETOOTH_SCAN"/>

    <!-- Location Permission -->
    <uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"/>

    <!-- Vibrator Permission -->
    <uses-permission android:name="android.permission.VIBRATE"/>

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Material3.DayNight.NoActionBar">

        <activity
            android:name=".MainActivity"
            android:exported="true">

```

```
<intent-filter>  
    <action android:name="android.intent.action.MAIN"/>  
  
    <category  
        android:name="android.intent.category.LAUNCHER"/>  
</intent-filter>
```

```
</activity>
```

```
</application>
```

```
</manifest>
```

Output:



9. Create an android application to get the notifications on Notification Bar by Using Notification Service.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <Button
        android:id="@+id/btnNotify"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Show Notification" />

</LinearLayout>
```

MainActivity.java

```
package com.example.BIHE;

import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.os.Build;
import android.os.Bundle;
import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.NotificationCompat;
import androidx.core.app.NotificationManagerCompat;

public class MainActivity extends AppCompatActivity {
```

```
Button btnNotify;
```

```
@Override
```

```
protected void onCreate(Bundle savedInstanceState) {
```

```
    super.onCreate(savedInstanceState);
```

```
    setContentView(R.layout.activity_main);
```

```
    btnNotify = findViewById(R.id.btnNotify);
```

```
    createNotificationChannel();
```

```
    btnNotify.setOnClickListener(v -> showNotification());
```

```
}
```

```
private void showNotification() {
```

```
    if (android.os.Build.VERSION.SDK_INT >=
```

```
        android.os.Build.VERSION_CODES.TIRAMISU) {
```

```
        if (checkSelfPermission(
```

```
            android.Manifest.permission.POST_NOTIFICATIONS)
```

```
            != android.content.pm.PackageManager.PERMISSION_GRANTED) {
```

```
            requestPermissions(
```

```
                new String[]{
```

```
                    android.Manifest.permission.POST_NOTIFICATIONS
```

```
                },
```

```
                100);
```

```
            return;
```

```
        }
```

```
    }
```

```

NotificationCompat.Builder builder =
    new NotificationCompat.Builder(this, "my_channel")
        .setSmallIcon(android.R.drawable.ic_dialog_info)
        .setContentTitle("Notification Service")
        .setContentText(
            "Welcome to Android Notification Service")
        .setPriority(
            NotificationCompat.PRIORITY_DEFAULT);

NotificationManagerCompat notificationManager =
    NotificationManagerCompat.from(this);

notificationManager.notify(1, builder.build());
}

private void createNotificationChannel() {
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {

        NotificationChannel channel =
            new NotificationChannel(
                "my_channel",
                "My Notifications",
                NotificationManager.IMPORTANCE_DEFAULT);

        channel.setDescription("Notification Channel");

        NotificationManager manager =
            getSystemService(NotificationManager.class);

        manager.createNotificationChannel(channel);
    }
}

```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <uses-permission
        android:name="android.permission.POST_NOTIFICATIONS" />

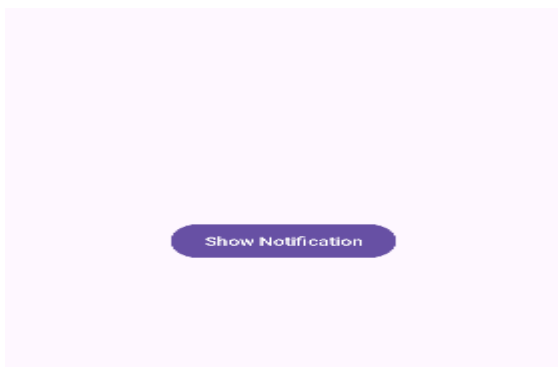
    <application
        android:allowBackup="true"
        android:label="@string/app_name"
        android:supportsRtl="true"
        android:theme="@style/Theme.Material3.DayNight.NoActionBar">
        <activity
            android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN" />
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>

        </activity>

    </application>

</manifest>
```

Output:



10. Create an android application to display available Wi-Fi devices and Paired Wi-Fi devices by using Wi-Fi Service.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="10dp">

    <Button
        android:id="@+id/btnScan"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Scan Wi-Fi Networks"/>

    <ListView
        android:id="@+id/listWifi"
        android:layout_width="match_parent"
        android:layout_height="match_parent"/>
```

```
</LinearLayout>
```

MainActivity.java

```
package com.example.BIHE;

import static com.example.BIHE.R.*;
import android.Manifest;
import android.annotation.SuppressLint;
import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.content.IntentFilter;
```

```
import android.content.pm.PackageManager;
import android.net.wifi.ScanResult;
import android.net.wifi.WifiManager;
import android.os.Build;
import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.ListView;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.content.ContextCompat;

import java.util.ArrayList;
import java.util.List;

public class MainActivity extends AppCompatActivity {

    private WifiManager wifiManager;
    private Button btnScan;
    private ListView listWifi;

    private ArrayList<String> wifiNetworks;
    private ArrayAdapter<String> adapter;

    private static final int REQUEST_CODE = 100;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

```

btnScan = findViewById(R.id.btnScan);
listWifi = findViewById(R.id.listWifi);

wifiManager = (WifiManager)
    getApplicationContext()
        .getSystemService(Context.WIFI_SERVICE);

wifiNetworks = new ArrayList<>();

adapter = new ArrayAdapter<>(
    this,
    android.R.layout.simple_list_item_1,
    wifiNetworks);

listWifi.setAdapter(adapter);

requestPermissionsIfNeeded();

btnScan.setOnClickListener(v -> scanWifi());

registerReceiver(
    wifiReceiver,
    new IntentFilter(
        WifiManager.SCAN_RESULTS_AVAILABLE_ACTION));
}

private void requestPermissionsIfNeeded() {

    if (Build.VERSION.SDK_INT >=
        Build.VERSION_CODES.TIRAMISU) {

```

```

if (ContextCompat.checkSelfPermission(
    this,
    Manifest.permission.NEARBY_WIFI_DEVICES)
    != PackageManager.PERMISSION_GRANTED) {

    ActivityCompat.requestPermissions(
        this,
        new String[]{
            Manifest.permission.NEARBY_WIFI_DEVICES
        },
        REQUEST_CODE);
}

} else {

    if (ContextCompat.checkSelfPermission(
        this,
        Manifest.permission.ACCESS_FINE_LOCATION)
        != PackageManager.PERMISSION_GRANTED
        || ContextCompat.checkSelfPermission(
            this,
            Manifest.permission.ACCESS_COARSE_LOCATION)
            != PackageManager.PERMISSION_GRANTED) {

        ActivityCompat.requestPermissions(
            this,
            new String[]{
                Manifest.permission.ACCESS_COARSE_LOCATION,
                Manifest.permission.ACCESS_FINE_LOCATION
            },
            REQUEST_CODE);
    }
}

```

```

    }
}

@SuppressLint("MissingPermission")
private void scanWifi() {

    try {

        if (Build.VERSION.SDK_INT >=
Build.VERSION_CODES.TIRAMISU) {

            if (ContextCompat.checkSelfPermission(
                this,
                Manifest.permission.NEARBY_WIFI_DEVICES)
                != PackageManager.PERMISSION_GRANTED) {

                Toast.makeText(
                    this,
                    "Wi-Fi permission required",
                    Toast.LENGTH_SHORT).show();
                return;
            }

        } else {

            if (ContextCompat.checkSelfPermission(
                this,
                Manifest.permission.ACCESS_FINE_LOCATION)
                != PackageManager.PERMISSION_GRANTED
                || ContextCompat.checkSelfPermission(
                    this,
                    Manifest.permission.ACCESS_COARSE_LOCATION)
                != PackageManager.PERMISSION_GRANTED) {

```

```

        Toast.makeText(
            this,
            "Location permission required",
            Toast.LENGTH_SHORT).show();
        return;
    }
}

boolean success = wifiManager.startScan();

if (!success) {

    Toast.makeText(
        this,
        "Wi-Fi Scan Failed",
        Toast.LENGTH_SHORT).show();
    }

} catch (SecurityException e) {

    Toast.makeText(
        this,
        "Permission denied",
        Toast.LENGTH_SHORT).show();
    }
}

private final BroadcastReceiver wifiReceiver =
    new BroadcastReceiver() {

        @SuppressWarnings("MissingPermission")
        @Override

```

```

public void onReceive(
    Context context,
    Intent intent) {

    try {

        List<ScanResult> results =
            wifiManager.getScanResults();

        wifiNetworks.clear();

        for (ScanResult result : results) {

            wifiNetworks.add(
                "SSID : " + result.SSID +
                "\nBSSID : " + result.BSSID +
                "\nSignal : " + result.level + " dBm");
        }

        adapter.notifyDataSetChanged();

    } catch (SecurityException e) {

        Toast.makeText(
            context,
            "Unable to read Wi-Fi networks",
            Toast.LENGTH_SHORT).show();
    }
}
};

```

@Override

```
public void onRequestPermissionsResult(
```

```
int requestCode,
@NonNull String[] permissions,
@NonNull int[] grantResults) {

super.onRequestPermissionsResult(
    requestCode,
    permissions,
    grantResults);

if (requestCode == REQUEST_CODE) {

    if (grantResults.length > 0
        && grantResults[0]
        == PackageManager.PERMISSION_GRANTED) {

        Toast.makeText(
            this,
            "Permission Granted",
            Toast.LENGTH_SHORT).show();

    } else {

        Toast.makeText(
            this,
            "Permission Denied",
            Toast.LENGTH_SHORT).show();
    }
}

}

@Override
protected void onDestroy() {
```

```

super.onDestroy();

try {
    unregisterReceiver(wifiReceiver);
} catch (Exception ignored) {
}
}
}
}

AndroidManifest.xml
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android">

    <uses-permission
android:name="android.permission.ACCESS_WIFI_STATE"/>
    <uses-permission
android:name="android.permission.CHANGE_WIFI_STATE"/>

    <uses-permission
android:name="android.permission.ACCESS_COARSE_LOCATION"/>
    <uses-permission
android:name="android.permission.ACCESS_FINE_LOCATION"/>

    <uses-permission
        android:name="android.permission.NEARBY_WIFI_DEVICES"
        android:usesPermissionFlags="neverForLocation"/>

    <application
        android:allowBackup="true"
        android:label="WiFi Scanner"
        android:theme="@style/Theme.AppCompat.Light">

        <activity
            android:name=".MainActivity"

```

```
    android:exported="true">

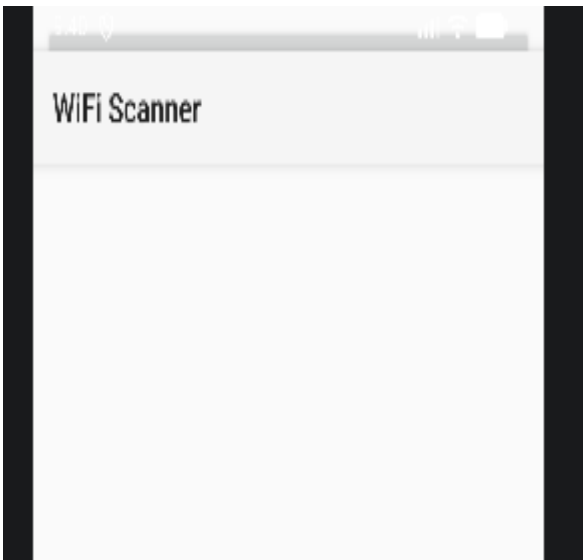
    <intent-filter>
        <action android:name="android.intent.action.MAIN"/>
        <category
android:name="android.intent.category.LAUNCHER"/>
    </intent-filter>

</activity>

</application>

</manifest>
```

Output:



11. Create an android application to display X, Y Sensor values by using Sensor Service.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerHorizontal="true"
        android:layout_marginTop="120dp"
        android:text="X Value: 0.0"
        android:textSize="25sp" />

    <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerInParent="true"
        android:text="Y Value: 0.0"
        android:textSize="25sp" />

</RelativeLayout>
```

MainActivity.java

```
package com.example.BIHE;

import static com.example.BIHE.R.*;

import android.content.Context;
import android.hardware.Sensor;
import android.hardware.SensorEvent;
import android.hardware.SensorEventListener;
import android.hardware.SensorManager;
import android.os.Bundle;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity implements
SensorEventListener {

    TextView tv1, tv2;
    SensorManager sensorManager;
    Sensor accelerometer;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        tv1 = findViewById(R.id.textView);
        tv2 = findViewById(R.id.textView2);

        // Get Sensor Service
        sensorManager = (SensorManager)
getSystemService(Context.SENSOR_SERVICE);
```

```

        if (sensorManager != null) {
            accelerometer =
sensorManager.getDefaultSensor(Sensor.TYPE_ACCELEROMETER);
        }
    }
}

```

```

@Override
protected void onResume() {
    super.onResume();

    if (accelerometer != null) {
        sensorManager.registerListener(
            this,
            accelerometer,
            SensorManager.SENSOR_DELAY_NORMAL
        );
    }
}

```

```

@Override
protected void onPause() {
    super.onPause();
    sensorManager.unregisterListener(this);
}

```

```

@Override
public void onSensorChanged(SensorEvent event) {

    if (event.sensor.getType() == Sensor.TYPE_ACCELEROMETER)
    {

        float x = event.values[0];
    }
}

```

```
float y = event.values[1];
```

```
tv1.setText("X Value is: " + x);
```

```
tv2.setText("Y Value is: " + y);
```

```
}
```

```
}
```

```
@Override
```

```
public void onAccuracyChanged(Sensor sensor, int accuracy) {
```

```
    // Not required
```

```
}
```

```
}
```

Output:



X Value is: 0.0

Y Value is: 9.809989

12. Create an android application to get the System Announcements by using Broadcast Receiver.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <TextView
        android:id="@+id/tv1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Waiting for System Broadcast..."
        android:textSize="24sp"
        android:textColor="#000000"
        android:layout_centerInParent="true"/>

</RelativeLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.BIHE">

    <application
        android:allowBackup="true"
        android:label="Broadcast Receiver App"
        android:theme="@style/Theme.AppCompat.Light.NoActionBar">

        <activity android:name=".MainActivity"
            android:exported="true">
            <intent-filter>
                <action android:name="android.intent.action.MAIN"/>
            </intent-filter>
        </activity>
    </application>
</manifest>
```

```

        <category android:name="android.intent.category.LAUNCHER"/>
    </intent-filter>
</activity>

```

```

</application>

```

```

</manifest>

```

MainActivity.java

```

package com.example.BIHE;

import android.content.Intent;
import android.content.IntentFilter;
import android.os.Bundle;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    TextView tv;
    MyReceiver receiver;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        tv = findViewById(R.id.tv1);

        receiver = new MyReceiver(tv);

        IntentFilter filter = new IntentFilter();
        filter.addAction(Intent.ACTION_POWER_CONNECTED);
    }
}

```

```

filter.addAction(Intent.ACTION_POWER_DISCONNECTED);
filter.addAction(Intent.ACTION_SCREEN_ON);
filter.addAction(Intent.ACTION_SCREEN_OFF);
filter.addAction(Intent.ACTION_AIRPLANE_MODE_CHANGED);
filter.addAction(Intent.ACTION_HEADSET_PLUG);

registerReceiver(receiver, filter);
}

```

@Override

```

protected void onDestroy() {
    super.onDestroy();
    unregisterReceiver(receiver);
}
}

```

MyReceiver.java

```

package com.example.BIHE;

import android.content.BroadcastReceiver;
import android.content.Context;
import android.content.Intent;
import android.widget.TextView;

public class MyReceiver extends BroadcastReceiver {

    TextView tv;

    public MyReceiver(TextView tv) {
        this.tv = tv;
    }

    @Override
    public void onReceive(Context context, Intent intent) {

```

```
String action = intent.getAction();

if (action == null || tv == null) return;

switch (action) {

    case Intent.ACTION_POWER_CONNECTED:
        tv.setText("POWER CONNECTED");
        break;

    case Intent.ACTION_POWER_DISCONNECTED:
        tv.setText("POWER DISCONNECTED");
        break;

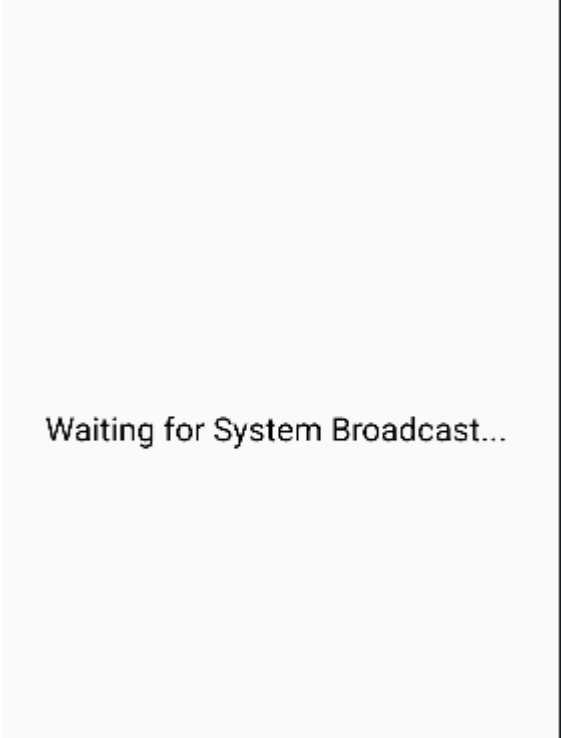
    case Intent.ACTION_SCREEN_ON:
        tv.setText("SCREEN ON");
        break;

    case Intent.ACTION_SCREEN_OFF:
        tv.setText("SCREEN OFF");
        break;

    case Intent.ACTION_AIRPLANE_MODE_CHANGED:
        tv.setText("AIRPLANE MODE CHANGED");
        break;

    case Intent.ACTION_HEADSET_PLUG:
        tv.setText("HEADSET PLUGGED");
        break;
}
}
```

Output:



Waiting for System Broadcast...